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Recent trends in thermal atomic layer deposition chemistry

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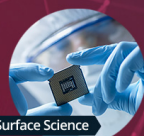
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Recent trends in thermal atomic layer deposition chemistry

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ABSTRACT

In this review, we highlight new atomic layer deposition (ALD) precursors and process chemistries based on the ALD database found in atomicleimits.com. The aim was to compare the processes before and after 2010 and see possible changes. The motivations for process development and trends in the types of different metal precursors are discussed. The total number of published thermal ALD processes is 1711, of which more than half (942) were published after 2010. The number of materials deposited by thermal ALD is 539, and for 312 of these, the process was published after 2010. The most popular material group are binary oxides. After 2010, the share of nonoxide and ternary materials slowly increased. During the last years, a few material classes have come forth, viz., metals, 2D transition metal dichalcogenides, and halides. The development of new ALD processes is clearly application-driven and visible in these material classes, motivated by the most important application areas of ALD: Microelectronics, energy technology, and catalysis. New elements added to the portfolio after 2010 are alkali metals (Na, K, and Rb), Be, Re, Os, Au, and Sb, the first two as oxides and the latter four as metals. The processes for Re, Os, Au, and Sb were different: Reductive for Re, oxidative for Os and Au, and exchange reaction for Sb. ALD of transition metals has been of interest because of their potential use in microelectronics. New metal precursors and novel reducing agents play an important role in their process development. Metal halides, alkoxides, alkyl compounds, β -diketonates, and amides/imides have been traditional metal precursors in ALD. After 2010, amides/imides have been the most applied precursors in new ALD processes, followed by cyclopentadienyl compounds. However, heteroleptic complexes containing two or more ligands are the largest precursor type, and they usually consist of a mixture of the above-mentioned ligands. The use of heteroleptic compounds enables tuning of precursor properties such as volatility, reactivity, and stability.

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I. INTRODUCTION

Atomic layer deposition (ALD) is a chemical gas-phase thin film deposition technique, which has similarities with the more well-known chemical vapor deposition (CVD). In contrast to CVD, where the gaseous precursors are introduced simultaneously into the reactor in ALD, the chemical reactions are split into two or more reacting steps and repeated in a cyclic manner.¹ The reactor is purged with an inert gas pulse between the precursor pulses. Precursors react only with the functional groups on the substrate surface, and the reactions are self-limiting, reaching saturation after all the surface sites are occupied. This self-limiting growth brings several unique advantages, such as precise control of film thickness

at the atomic level, excellent uniformity across large surface areas, and excellent conformality in complex 3D structures with high aspect ratios.^{2–5}

ALD technology relies on surface chemistry, and a complete and fast reaction of precursors is essential. The combination of precursors is important because it influences the properties of the deposited films, such as film thickness, morphology, conformality, and composition. ALD has over 50 years history,^{6–8} but its development was initially quite slow. It accelerated significantly in the late 1990s when it became clear that following Moore's law in microelectronics is impossible without new materials and thin film deposition techniques. Today, the microelectronics industry is in nanometer or

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Ångström era as well as three-dimensional and, therefore, requires the use of ALD in the most critical process steps.⁹

As an embodiment of the research activity, ALD literature has increased rapidly since the late 1990s. The annual number of ALD papers is approximately 1800, but most of them deal with the applications of ALD materials. The number of papers reporting new ALD processes is significantly lower—annually 60–80. In their 2013 review paper, Miikkulainen *et al.* listed the ALD processes known at the end of 2011.¹⁰ Since then, many new materials have been deposited by ALD, and new processes have been developed for earlier known ALD materials. Recently, all the ALD processes reported have been collected in one place, the ALD database (Fig. 1), and researchers are completing the collection when new processes appear.¹¹ Thanks to this practical source of literature, there is no need for printed reviews or tables of ALD processes covering all the processes from the beginning but focus can be on the recent processes.

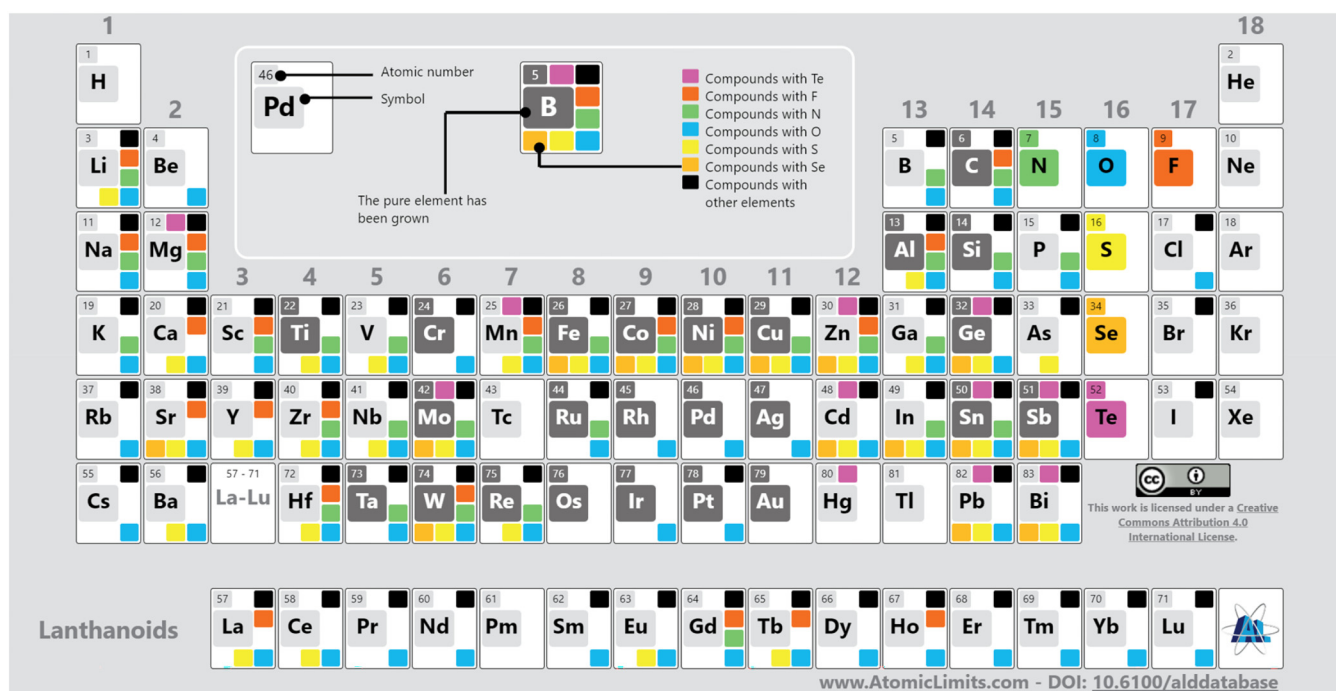
This review focuses on thermal ALD, which is a completely precursor-dependent method. The success of the process is based on a favorable combination of precursors that should react completely and quickly in the alternative surface reactions. The most prominent advantage of thermal ALD is its ability to provide perfect conformal coverage in high aspect ratio features. Plasma-enhanced ALD is another main ALD technology where the surface reactions are affected by external energy. The use of plasma species as reactants allows for milder processing conditions and the preparation of materials that are not possible with the conventional thermally driven ALD method. However, conformal coverage in

high aspect ratio features can be poor in plasma ALD due to the recombination of plasma species on the feature walls before reaching the bottom. In addition, controlling the process can be more complicated because of the new variables connected to the plasma generator. For more information on plasma ALD, we refer the reader to review articles by Profijt *et al.*¹² and Knoops *et al.*¹³

II. METHODS

The data snapshot was taken from the AtomicLimits ALD database¹¹ on 13.01.2024 (3022 entries). The database is licensed under CC BY 4.0, see <https://creativecommons.org/licenses/by/4.0/>. At every step of data processing, a pre- and postcopy of the database was kept to allow manual checking, traceback, and error correction. Data processing included automated and manual steps.

The automated data processing steps were as follows: Plasma processes were filtered out by keyword “plasma” in the precursor fields leaving 2432 entries. Duplicates were filtered out by a search for entries with the same combination (C) of elements in the material deposited and precursor fields, leaving 1715 process entries. Manual review of these entries led to the removal of three MLD processes and one entry where a different precursor abbreviation caused duplication. The final number of thermal process entries was 1711. Visual Basic macro for Excel was used to extract the DOIs of the references from the references field. In the case of multiple references, the oldest reference by publication date was used.



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FIG. 1. Periodic table of ALD processes generated by the ALD database (www.atomiclimits.com/alddbbase). Generated on 02.02.2025. This table includes both thermal and plasma ALD processes.

Bash scripts were used to fetch the bibliographic data assigned to the DOIs and append them to their respective entry fields.

Owing to the inconsistencies in the way the ALD database entries are filed, automation proved unreliable for identifying how many elements the deposited material contains, whether it is unique (for example, Cu_2O versus CuO versus CuO_x) or appending precursor-based tags. This data processing was performed by hand, and the corresponding reference was checked if necessary.

For Fig. 17, the data were obtained using the Web of Science™ on 12.10.2022 with the following search queries and refining keywords: Solar cells—“halide perovskite solar cell” refined with “tandem,” “flexible” or “lead free;” light emitting diodes—“halide perovskite light emitting diode;” lasers—“halide perovskite laser,” sensing—“halide perovskite detector OR halide perovskite sensor OR halide perovskite scintillator OR halide perovskite x-ray imaging OR halide perovskite x-ray detection OR halide perovskite scintillator OR halide perovskite x-ray imaging OR” refined with “x-ray;” microelectronics—“halide perovskite microelectronics OR halide perovskite transistor OR halide perovskite memory” refined with “memory” or “transistor;” energy storage—“halide perovskite energy storage OR halide perovskite battery” refined with “lithium;” photocatalysis—“halide perovskite photocatalysis” refined with “ CO_2 ,” thermal energy—“halide perovskite pyroelectric OR halide perovskite thermoelectric” refined with “thermoelectric;” mechanical energy—“halide perovskite piezoelectric OR halide perovskite triboelectric” refined with “piezoelectric.”

III. ALD PROCESSES AND MATERIALS

A. Processes

The database contains 1711 unique thermal ALD processes for 539 thin film materials published between 1975 and 2023. The boom in ALD started around 1995, and the number of new processes reported annually increased from ~20 in the early 1990s to 60 in 2008 (Fig. 2). Since then, the number of new processes has remained at 60–80 each year. Since 2010, 942 new processes have been published for 431 materials, of which 312 materials did not earlier have any ALD process (Table I). The numbers show high recent activity since 55% of all processes and 58% of all materials were published after 2010.

The slope in the number of processes versus time shows a small decrease, indicating saturation in the activity for the new ALD processes. One reason for this “saturation” may be that most important thin film materials already have applicable processes. Process development is reasoned by two factors: A material does not have any ALD process, or the reported processes have shortcomings that limit their universal applications. Development of better ALD chemistry and cheaper precursors is an everlasting process. The development of processes for new materials is driven either by curiosity or by the need in certain applications.

B. ALD materials based on the number of elements

Taking into account all ALD processes, 57% of the thin film materials are binary compounds (Fig. 3). The count for ternaries is close to 30%, 10% for elements, and 4% for quaternaries. We note that ternaries (quaternaries) may be true ternary (quaternary)

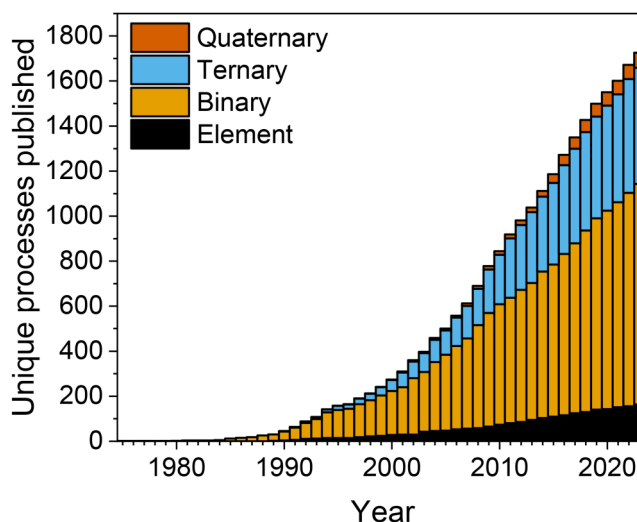


FIG. 2. Cumulative number of new thermal ALD processes published during 1975–2010. The different colors show the different types of materials based on the number of elements in the materials.

compounds with a fixed stoichiometry or mixtures of two (three) binary compounds. When comparing the numbers before and after 2010, the share of binary compounds decreased from 65% to 50% and that of ternaries increased from 24% to 34%. The share of elements and quaternary materials has increased since 2010. It is quite natural that the processes have been studied first for easier systems, such as binaries. The deposition of multicomponent films is more demanding than that of binary compounds because, for example, the growth rate may be different in multicomponent processes than in binary processes, making stoichiometry control challenging. Recent interest in ALD metal films can be seen in the increased number of ALD processes for elements.

The curves showing the number of processes for different material types (Fig. 4) contain random variations from year to year. In the total number of processes, there was a steep increase until 2010, after which it stabilized at the level of 60–80 per year. Interestingly, it appears that the number of binary processes follows that of all processes, increasing to 2010 and then stabilizing with annual variations.

Ternary materials appeared in the ALD literature much later than binary compounds. Their development grew in the late 1990s and peaked in 2011. After that, the number slowly decreased. The six elements with the most ternary material processes are Al (103), Ti (89), Si (57), Zn (57), Hf (52), and Zr (45). The interest in the latter two stems from their use as high- k dielectrics, which has declined over time. Al, Ti, and Zn are flagship ALD elements with either exemplary precursors available such as TMA, DEZ, or vast selection of well-studied precursors such as halides, alkoxides, and amides for Ti. For these elements, the decrease in annual published ternary processes is due to the options being slowly exhausted over time. The interest in Si is due to silicates being more easily accessible than binary Si compounds.⁵⁰

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TABLE I. List of thermal processes that were published in 2010–2023 as well as submitted and accepted to the ALD database (Ref. 11) before 13.01.2024.

Material	Reactant A	Reactant B	Reactant C	Further reactants	First Ref
Ag	Ag(hfac)(cod)	Propanol			Ref. 14
Ag	Ag(hfac)(PMe ₃)	Formalin			Ref. 15
Ag	Ag(hfac)(PMe ₃)	TMA	H ₂ O		Ref. 15
Ag	Ag(fod)(PEt ₃)	BH ₃ (NHMe ₂)			Ref. 16
Ag	Ag(hfac)(cod)	^t BuNHNH ₂			Ref. 17
Ag ₂ O	Ag(hfac)(PMe ₃)	O ₃			Ref. 18
Al	AlCl ₃	AlH ₂ (^t BuN)CH ₂ CH ₂ (NMe ₂)			Ref. 19
Al ₂ O ₃	AlCl ₃	H ₂ O + catalyst			Ref. 20
Al ₂ O ₃	AlCl ₃	O ₃			Ref. 21
Al ₂ O ₃	Al(NMe ₂) ₃	H ₂ O			Ref. 22
Al ₂ O ₃	Al(N ⁱ Pr) ₂ (C ₃ H ₆ NMe ₂)	H ₂ O			Ref. 23
Al ₂ O ₃	AlMe ₂ Cl	O ₃			Ref. 24
Al ₂ O ₃	Al(O ^s Bu) ₃	H ₂ O			Ref. 25
Al ₂ O ₃	Al(O ⁱ Pr) ₃	H ₂ O			Ref. 26
Al ₂ O ₃	AlMe ₃	EtOH			Ref. 27
Al ₂ O ₃	AlMe ₃	MeOH			Ref. 27
Al ₂ O ₃	AlMe ₃	H ₂			Ref. 28
Al ₂ O ₃	AlMe ₃	H ₂ O	O ₃		Ref. 29
Al ₂ S ₃	AlMe ₃	H ₂ S			Ref. 30
Al ₂ S ₃	Al(NMe ₂) ₃	H ₂ S			Ref. 31
AlF ₃	AlCl ₃	TiF ₄			Ref. 32
AlF ₃	AlMe ₃	HF			Ref. 33
AlF ₃	AlMe ₃	TaF ₅			Ref. 34
AlF ₃	Al(NMe ₂) ₃	HF			Ref. 34
AlGa _x O _y	AlMe ₃	GaMe ₃	O ₃ + H ₂ O		Ref. 35
AlMo _x O _y	AlMe ₃	MoCl ₄ O	H ₂ O		Ref. 36
AlN	Al(NEt ₂) ₃	NH ₃			Ref. 37
AlN	Al(NEt ₂) ₃	N ₂ H ₄			Ref. 37
AlN	Al(ⁱ Bu) ₃	N ₂ H ₄ ·HCl			Ref. 38
AlN	AlH ₂ (NMe ₂)	NH ₃			Ref. 39
AlN	AlMe ₃	MeHNNH ₂			Ref. 40
AlO _x F _y	AlMe ₃	H ₂ O	HF		Ref. 41
AlO _x N _y	Al ₂ (NMe ₂) ₆	NH ₃	H ₂ O		Ref. 42
AlP _x O _y	AlCl ₃	PO(OMe) ₃	H ₂ O	O ₃	Ref. 43
AlP _x O _y	AlCl ₃	PO(OEt) ₃			Ref. 44
AlP _x O _y	AlMe ₃	P(NMe ₂) ₃	H ₂ O		Ref. 45
AlP _x O _y	AlMe ₃	PO(OMe) ₃	H ₂ O		Ref. 46
AlP _x O _y	AlMe ₃	P(OMe) ₃	H ₂ O		Ref. 47
AlSb	AlCl ₃	(Et ₃ Si) ₃ Sb			Ref. 48
AlSc _x O _y	AlMe ₃	Sc(MeCp) ₃	H ₂ O		Ref. 49
AlSi _x O _y	AlCl ₃	Si(O ^t Bu) ₄			Ref. 50
AlSi _x O _y	AlCl ₃	Si(O ⁱ Bu) ₄	H ₂ O		Ref. 50
AlSi _x O _y	AlCl ₃	Si(OEt) ₄	H ₂ O		Ref. 50
AlSi _x O _y	AlCl ₃	Si(O ^t Bu) ₃ OH			Ref. 50
AlSi _x O _y	AlCl ₃	Si(O ⁱ Bu) ₃ OH	H ₂ O		Ref. 50
AlSi _x O _y	AlMe ₂ Cl	Si(O ^t Bu) ₄	H ₂ O		Ref. 50
AlSi _x O _y	AlMe ₂ Cl	Si(OEt) ₄	H ₂ O		Ref. 50
AlSi _x O _y	AlMe ₃	Si(O ^t Bu) ₄	H ₂ O		Ref. 50
AlSi _x O _y	AlMe ₃	Si(OEt) ₄	H ₂ O		Ref. 50
AlSi _x O _y	AlMe ₃	Si(O ^t Bu) ₃ OH			Ref. 50
AlSi _x O _y	AlMe ₃	Si(O ⁱ Bu) ₃ OH	H ₂ O		Ref. 50
AlSi _x O _y	AlMe ₃	Si ₂ (NHet) ₆	D ₂ O		Ref. 51

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TABLE I. (Continued.)

Material	Reactant A	Reactant B	Reactant C	Further reactants	First Ref
AlSi _x O _y	AlMe ₃	Si(OEt) ₃ (ⁿ PrNH ₂)	H ₂ O	H ₂ O + O ₃	Ref. 52
AlW _x F _y	WF ₆	AlMe ₃			Ref. 53
AlZrO	Al(CH ₃) ₃	(MeCp) ₂ ZrMe(OMe)	H ₂ O		Ref. 54
As ₂ S ₃	As(NMe ₂) ₃	H ₂ S			Ref. 55
Au	Au(N(SiMe ₃) ₂)(PEt ₃)	BH ₃ (NHMe ₂)			Ref. 56
Au	Me ₂ Au(S ₂ CNEt ₂)	O ₃			Ref. 57
Au	Au(PMe ₃)Me ₃	O ₃	H ₂ O		Ref. 58
Au	AuCl(PEt ₃)	(Me ₃ Ge) ₂ DHP			Ref. 59
B ₂ O ₃	B(OMe) ₃	O ₃			Ref. 60
B ₂ O ₃	BCl ₃	H ₂ O			Ref. 61
BaSnO ₃	Ba(ⁱ Pr ₃ Cp) ₂	Sn(NMe ₂) ₄	H ₂ O		Ref. 62
BaTi _x O _y	Ba(Cp ⁱ Pr ₃) ₂	Ti(O ⁱ Pr) ₄	H ₂ O		Ref. 63
BaTi _x O _y	Proprietary py-Ba precursor	Ti(O ⁱ Pr) ₄	H ₂ O		Ref. 64
BeO	BeMe ₂	H ₂ O			Ref. 65
BeO	BeEt ₂	H ₂ O			Ref. 66
BeO	BeMe ₂	O ₃			Ref. 67
Bi ₂ O ₃	Bi(OCMe ₂ ⁱ Pr) ₃	H ₂ O			Ref. 68
Bi ₂ S ₃	Bi(thd) ₃	H ₂ S			Ref. 69
Bi ₂ Se ₃	BiCl ₃	H ₂ Se			Ref. 70
Bi ₂ Te ₃	[(SiMe ₃) ₂ NBi-μ-NSiMe ₃] ₂	Te(SiEt ₃) ₂			Ref. 71
Bi ₂ Te ₃	Bi(NMe ₂) ₃	Te(SiEt ₃) ₂			Ref. 71
Bi ₂ Te ₃	Bi(NMeEt) ₃	Te(SiEt ₃) ₂			Ref. 71
BiFe _x O _y	Bi(mmp) ₃	FeCp ₂	O ₃		Ref. 72
BiFe _x O _y	Bi(thd) ₃	Fe(thd) ₃	H ₂ O		Ref. 73
BiFe _x O _y	BiMe ₃	C ₆ H ₈ Fe(CO) ₃	H ₂ O		Ref. 74
BiFe _x O _y	Bi(OC ₆ H ₁₃) ₃	Fe(O ⁱ Bu) ₃	H ₂ O		Ref. 75
BiFe _x O _y	Bi(thd) ₃	FeCp ₂	H ₂ O	O ₃	Ref. 76
BiFe _x O _y	BiPh ₃	FeCp ₂	O ₃		Ref. 77
BiOCl	BiCl ₃	O ₃			Ref. 78
BiO _x	Bi(dmb) ₃	H ₂ O			Ref. 79
BiO _x	Bi(O ⁱ Bu) ₃	H ₂ O			Ref. 79
BiO _x	Bi(thd) ₃	H ₂ O			Ref. 79
BiSe	Bi(NMe ₂) ₃	Se(SnMe ₃) ₂			Ref. 80
BiV _x O _y	BiPh ₃	VO(O ⁱ Pr) ₃	O ₃	H ₂ O	Ref. 81
BiV _x O _y	BiPh ₃	VO(O ⁱ Pr) ₃	H ₂ O		Ref. 81
BiV _x O _y	BiPh ₃	VO(O ⁱ Pr) ₃	H ₂ O	ROH	Ref. 82
BN	B ₃ Cl ₃ H ₃ N ₃	HN[SiMe ₃] ₂			Ref. 83
BN	B ₃ N ₃ H ₆	Electrons			Ref. 84
[Ca ₂ CoO ₃] _{0.6} •2[CoO ₂]	Ca(thd) ₂	Co(thd) ₂	O ₃		Ref. 85
CaB ₂ O ₄	CaTp ₂	H ₂ O			Ref. 86
CaC _x O _y	Ca(thd) ₂	O ₃			Ref. 87
CaC _x O _y	Ca(thd) ₂	CO ₂	O ₃		Ref. 87
CaF ₂	Ca(hfac) ₂	O ₃			Ref. 88
CaF ₂	Ca(thd) ₂	O ₃	Hhfac	O ₃	Ref. 88
CaF ₂	Ca(ⁱ Pr ₂ FMD) ₂	HF			Ref. 89
CaMoO ₄	Ca(thd) ₂	O ₃	MoCl ₄ O	H ₂ O + O ₃	Ref. 90
CaS	Ca(ⁱ PrFMD) ₂	H ₂ S			Ref. 91
CaS:Eu	Ca(thd) ₂	Eu(thd) ₃	H ₂ S		Ref. 92
CaSnS	Ca(ⁱ PrFMD) ₂	Sn(ⁱ PrAMD) ₂	H ₂ S		Ref. 93
CdO	CdMe ₂	H ₂ O			Ref. 94
CdO _x S _y	CdCl ₂	H ₂ S	H ₂ O		Ref. 95

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TABLE I. (Continued.)

Material	Reactant A	Reactant B	Reactant C	Further reactants	First Ref
CdSe _x Te _y	Cd	Se	Te		Ref. 96
CdZn _x Ga _y S _z	CdMe ₂	ZnEt ₂	Ga(NMe ₂) ₃	H ₂ S	Ref. 97
CdZn _x O _y	CdMe ₂	ZnEt ₂	H ₂ O		Ref. 94
CdZn _x S _y	CdMe ₂	ZnEt ₂	H ₂ S		Ref. 98
CeHf _x O _y	Ce(Cp ⁱ Pr) ₃	Hf(NMe ₂) ₄	H ₂ O		Ref. 99
CeO ₂	Ce(ⁱ PrCp) ₃	H ₂ O			Ref. 99
CeO ₂	Ce(ⁱ PrCp) ₂ (ⁱ Pr ₂ AMD)	H ₂ O			Ref. 100
CeO ₂	Ce(ⁱ Pr ₂ AMD) ₃	O ₃			Ref. 101
CeO ₂	Ce(ⁱ Pr ₂ guan-NMe ₂) ₃	H ₂ O			Ref. 102
Co	Co(ⁱ PrAMD) ₂	NH ₃			Ref. 103
Co	^t Bu-AllylCo(CO) ₃	H ₂ NNMe ₂			Ref. 104
Co	Co(Me ⁱ PrCOCN ^t Bu) ₂	BH ₃ (NHMe ₂)			Ref. 105
Co	Co(^t Bu ₂ DAD) ₂	HCOOH			Ref. 106
Co	Co(^t Bu ₂ DAD) ₂	^t BuNH ₂			Ref. 107
Co	Co(^t Bu ₂ DAD) ₂	Et ₂ NH			Ref. 107
Co	Co(CO) ₃ NO	Electrons			Ref. 108
Co	CoCl ₂ (TMEDA)	Zn(DMP) ₂			Ref. 109
Co	Co(CpSiMe ₃)(CO) ₂	^t BuNH ₂			Ref. 110
Co	CoCp(CO) ₂	^t BuNH ₂			Ref. 110
Co ₃ O ₄	CoCp ₂	O ₃			Ref. 111
Co ₃ O ₄	Co(Cp)(CO) ₂	O ₃			Ref. 112
Co ₃ O ₄	Co(ⁱ Pr ₂ DAD) ₂	O ₃			Ref. 113
Co ₃ O ₄	Co(DMOCHCOCF ₃) ₂	O ₃			Ref. 114
Co ₃ O ₄	Co(^t Bu ₂ DAD) ₂	O ₃			Ref. 115
Co ₃ O ₄	Co(ⁱ Pr ₂ DAD) ₂	O ₂			Ref. 116
Co ₃ O ₄	Co(ⁱ PrAMD) ₂	O ₃			Ref. 117
Co ₃ Sn ₂	CoCl ₂ (TMEDA)	SnHBU ₃			Ref. 118
Co ₉ S ₈	Co(ⁱ PrMeAMD) ₂	H ₂ S			Ref. 119
Co ₉ S ₈	CoCl ₂ (TMEDA)	H ₂ S			Ref. 120
CoB ₂ O ₄	CoTp ₂	O ₃			Ref. 121
CoC _x	Co(acac) ₂	ⁿ PrOH			Ref. 122
CoF ₂	CoCl ₂ (TMEDA)	NH ₄ F			Ref. 123
CoFe _x O _y	CoCp ₂	FeCp ₂	O ₃		Ref. 124
CoFe _x O _y	CoCp ₂	FeCp ₂	O ₂		Ref. 125
CoFe _x O _y	Co(thd) ₂	Fe(thd) ₃	O radical		Ref. 126
CoO	Co(AMD) ₂	H ₂ O			Ref. 127
CoO	CoCl ₂ (TMEDA)	H ₂ O			Ref. 128
CoO	Co(N(SiMe ₃) ₂) ₂ (THF)	H ₂ O			Ref. 129
CoO	Co(ⁱ PrAMD) ₂	H ₂ O			Ref. 117
CoO/Co(OH) ₂	Co[N(SiMe ₃) ₂] ₂ (THF)	H ₂ O			Ref. 129
CoO _x	Co ₂ (CO) ₆ (^t BuC=CH)	O ₃			Ref. 130
CoO _x	Co ₂ (CO) ₈	O ₃			Ref. 131
CoO _x	Co(ⁱ Pr ₂ DAD) ₂	O ₂			Ref. 116
CoSi ₂	^t Bu-AllylCo(CO) ₃	Si ₃ H ₈			Ref. 132
CoTi _x N _y	Co(^t BuPAMD) ₂	Ti(NMe ₂) ₄	NH ₃		Ref. 133
Cr	Cr(Me ^t BuCOCN ^t Bu) ₂	BH ₃ (NHMe ₂)			Ref. 105
CrO _x	Cr(acac) ₃	O ₃			Ref. 134
CrO _x	Cr(thd) ₃	O ₃			Ref. 135
CsI	Cs(N(SiMe ₃) ₂)	SnI ₄			Ref. 136
CsNb _x O _y	Cs(O ^t Bu)	Nb(OEt) ₅	H ₂ O		Ref. 137
CsPbI ₃	Cs(N(SiMe ₃) ₂)	Pb(N(SiMe ₃) ₂) ₂	SnI ₄		Ref. 136

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TABLE I. (Continued.)

Material	Reactant A	Reactant B	Reactant C	Further reactants	First Ref
CsSnI ₃	Cs(N(SiMe ₃) ₂)	Sn(N(SiMe ₃) ₂) ₂	SnI ₄		Ref. 138
Cu	Cu(hfac) ₂	H ₂ + catalyst			Ref. 139
Cu	Cu(dmap) ₂	HCOOH	H ₂ NNH ₂		Ref. 140
Cu	Cu(dmap) ₂	BH ₃ (NHMe ₂)			Ref. 141
Cu	Cu(dmap) ₂	HCOOH	BH ₃ (NHMe ₂)		Ref. 141
Cu	Cu(ⁱ Pr ₂ NHC)(N(SiMe ₃) ₂)	H ₂			Ref. 142
Cu	Cu(hfac) ₂	ZnEt ₂			Ref. 143
Cu	Cu(acac) ₂	H ₂ O	C ₆ H ₄ (OH) ₂		Ref. 144
Cu	Cu(dmap) ₂	^t BuNHNH ₂			Ref. 145
Cu	Cu(hfac) ₂	H ₂ O			Ref. 146
Cu	Cu(acac) ₂	C ₆ H ₄ (OH) ₂			Ref. 147
Cu:ZnS _x	Cu(hfac) ₂	ZnEt ₂	H ₂ S		Ref. 148
Cu ₂ O	Cu(dmamb) ₂	H ₂ O			Ref. 149
Cu ₂ O	Cu(dmap) ₂	H ₂ O			Ref. 150
Cu ₂ O	[Cu(^s BuAMD)] ₂	H ₂ O			Ref. 151
Cu ₂ O	Cu(hfac) ₂	H ₂ O			Ref. 152
Cu ₂ O	Cu(OAc) ₂	H ₂ O			Ref. 153
Cu ₃ N	Cu(dmamb) ₂	NH ₃			Ref. 154
CuAl _x S _y	Cu(acac) ₂	AlMe ₃	H ₂ S		Ref. 155
CuCl	Cu(η ² -1,2-ethyne(SiMe ₃) ₂)(hfac)	HCl			Ref. 156
CuCr _x Mg _y O _z	Cu(thd) ₂	Cr(acac) ₃	Mg(thd) ₂	O ₃	Ref. 157
CuCr _x O _y	Cu(thd) ₂	Cr(acac) ₃	O ₃		Ref. 158
CuGa _x S _y	Cu(acac) ₂	Ga ₂ (NMe ₂) ₆	H ₂ S		Ref. 159
CuIn _x S _y	Cu(acac) ₂	In(acac) ₃	H ₂ S		Ref. 160
CuMn _x	Cu(dmap) ₂	[Mn(Me ^t BuCOCN ^t Bu) ₂] ₂	BH ₃ (NHMe ₂)		Ref. 161
CuO	Cu(acac) ₂	O ₃			Ref. 162
CuO	Cu(dmap) ₂	O ₃			Ref. 163
CuO	[Cu(^s BuAMD)] ₂	O ₃			Ref. 164
CuO	Cu(hfac)(dmb)	O ₃			Ref. 165
CuO	Cu(thd) ₂	O ₃			Ref. 166
CuSb _x S _y	[Cu(^s BuAMD)] ₂	Sb(NEt ₂) ₃	H ₂ S		Ref. 167
CuSn _x S _y	Cu(dmamb) ₂	Sn(dmamp) ₂	H ₂ S		Ref. 168
CuS _x	Cu(acac) ₂	H ₂ S			Ref. 155
CuS _x	Cu(acac) ₂	S			Ref. 169
CuS _x	Cu(dmamb) ₂	H ₂ S			Ref. 170
CuW _x O _y	[Cu(^s BuAMD)] ₂	W(^t BuN) ₂ (Me ₂ N) ₂	O ₃	H ₂ O	Ref. 164
CuZn _x	CuCl ₂	ZnEt ₂			Ref. 171
CuZn _x Sn _y S _z	[Cu(^s BuAMD)] ₂	ZnEt ₂	Sn(NMe ₂) ₄	H ₂ S	Ref. 172
CuZn _x S _y	[Cu(^s BuAMD)] ₂	ZnEt ₂	H ₂ S		Ref. 173
Dy ₂ O ₃	Dy(ⁱ Pr ₂ guan-NMe ₂) ₃	H ₂ O			Ref. 174
Dy ₂ O ₃	Dy(ⁱ PrCp) ₂ (ⁱ Pr ₂ AMD)	O ₃			Ref. 175
DySc _x O _y	Dy(thd) ₃	Sc(thd) ₃	O ₃		Ref. 176
DyTi _x O _y	Dy(thd) ₃	TiCl ₄	O ₃		Ref. 177
DyTi _x O _y	Dy(thd) ₃	TiCl ₄	O ₃	H ₂ O	Ref. 178
DyYb _x Ti _y O _z	Dy(thd) ₃	Yb(thd) ₃	TiCl ₄	O ₃ /H ₂ O	Ref. 178
DyZr _x O _y	Dy(thd) ₃	ZrCl ₄	H ₂ O		Ref. 179
Er ₂ O ₃	Er(CpMe) ₃	O ₃			Ref. 180
Er ₂ O ₃	Er(ⁱ Pr ₂ guan-NMe ₂) ₃	H ₂ O			Ref. 181
Er ₂ O ₃	Er(ⁱ PrCp) ₃	H ₂ O			Ref. 182
Er ₂ O ₃	Er(ⁱ PrCp) ₃	O ₃			Ref. 182
Er ₂ O ₃	Er(MeCp) ₂ (ⁱ PrAMD)	O ₃			Ref. 182

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TABLE I. (Continued.)

Material	Reactant A	Reactant B	Reactant C	Further reactants	First Ref
Er ₂ O ₃	Er(MeCp) ₂ (ⁱ PrAMD)	H ₂ O			Ref. 182
Er ₂ O ₃	Er(ⁿ BuCp) ₃	H ₂ O			Ref. 182
Er ₂ O ₃	Er(ⁿ BuCp) ₃	O ₃			Ref. 182
Er ₂ O ₃	Er(NMe ₂ CHCO ^t Bu) ₃	H ₂ O			Ref. 183
ErC ₂	Er(ⁱ PrCp) ₃	H ₂			Ref. 184
ErFe _x O _γ	Er(thd) ₃	Fe(thd) ₃	O ₃		Ref. 185
ErSc _x O _γ	Er(thd) ₃	Sc(thd) ₃	O ₃		Ref. 176
ErTi _x O _γ	Er(CpMe) ₃	Ti(NEt ₂) ₄	O ₃		Ref. 186
ErTi _x O _γ	Er(thd) ₃	TiCl ₄	O ₃	H ₂ O	Ref. 178
ErYb _x Ti _γ O _z	Er(thd) ₃	Yb(thd) ₃	TiCl ₄	O ₃ / H ₂ O	Ref. 178
ErY _x O _γ	Er(CpMe) ₃	YCp ₃	H ₂ O		Ref. 187
ErZr _x O _γ	Er(thd) ₃	Zr(CpMe) ₂ (OMe)Me	O ₃		Ref. 179
EuTi _x O _γ	Eu(thd) ₃	TiCl ₄	O ₃	H ₂ O	Ref. 188
EuTi _x P _γ O _z	Eu(thd) ₃	TiCl ₄	P(OMe) ₃ O	O ₃ /H ₂ O/ O ₃ +H ₂ O	Ref. 189
EuYb _x Ti _γ O _z	Eu(thd) ₃	Yb(thd) ₃	TiCl ₄	O ₃ /H ₂ O	Ref. 178
Fe	Fe(Me ^t BuCOCN ^t Bu) ₂	BH ₃ (NHMe ₂)			Ref. 105
FeAl _x O _γ	FeCp(^t BuCp)	O ₃	AlMe ₃	H ₂ O	Ref. 190
FeCo _x S _γ	Fe(ⁱ PrAMD) ₂	Co(ⁱ PrAMD) ₂	H ₂ S		Ref. 191
FeMg _x O _γ	FeCp(CpCH ₂ NMe ₂)	Mg(thd) ₂	O ₃		Ref. 192
FeMg _x O _γ	Fe ₂ (^t BuO) ₆	Mg(Cp) ₂	H ₂ O		Ref. 193
FeMg _x O _γ	FeCp ₂	Mg(thd) ₂	O ₃		Ref. 192
FeN _x	Fe(^t BuAMD) ₂	N ₂ H ₄			Ref. 194
FeO _x	Fe(acac) ₃	O ₃			Ref. 195
FeO _x	Fe(DMPD) ₂	O ₃			Ref. 196
FeO _x	Fe(DMPD) ₂	H ₂ O ₂			Ref. 196
FeO _x	Fe(hfac) ₂ (TMEDA)	O ₃			Ref. 197
FeO _x	Fe(N(SiMe ₃) ₂) ₂	H ₂ O ₂			Ref. 198
FeO _x	Fe(ipki) ₂	H ₂ O			Ref. 199
FeO _x	FeCl ₃	O ₃			Ref. 200
FeO _x	FeCp(^t BuCp)	O ₃			Ref. 201
FeP _x O _γ	Fe(thd) ₃	PO(OMe) ₃	O ₃	H ₂ O + O ₃	Ref. 202
FeP _x O _γ	FeCp ₂	PO(OMe) ₃	O ₃	H ₂ O	Ref. 203
FeRu _x O _γ	FeCl ₃	Ru(EtCp) ₂	O ₂ /H ₂ O		Ref. 204
FeSe _x	Fe(ⁱ PrAMD) ₂	Se(SiEt ₃) ₂			Ref. 205
FeSi _x O _γ	FeCl ₃	Si ₂ (NH ₂) ₆	O ₃		Ref. 200
FeS _x	Fe(^t BuAMD) ₂	H ₂ S			Ref. 206
FeTi _x O _γ	FeCp ₂	Ti(O ⁱ Pr) ₄	O ₃ +H ₂ O	H ₂ O	Ref. 207
FeTi _x O _γ	FeCp ₂	TiCl ₄	O ₃	H ₂ O	Ref. 208
FeZn _x	FeCl ₃	ZnEt ₂			Ref. 171
Ga ₂ O ₃	GaMe ₂ (O ⁱ Pr)	H ₂ O			Ref. 209
Ga ₂ O ₃	GaMe ₃	O ₃			Ref. 210
Ga ₂ O ₃	GaEt ₃	O ₃			Ref. 211
Ga ₂ O ₃	GaI ₃	O ₃			Ref. 212
GaAs	GaCl ₃	(Et ₃ Si) ₃ As			Ref. 213
GaSb	GaCl ₃	(Et ₃ Si) ₃ Sb			Ref. 48
GaS _x	Ga ₂ (NMe ₂) ₆	H ₂ S			Ref. 214
GaZn _x O _γ	Ga ₂ (NMe ₂) ₆	ZnEt ₂	H ₂ O		Ref. 215
GaZn _x O _γ	GaMe ₃	ZnEt ₂	O ₃		Ref. 216
Gd ₂ O ₃	Gd(ⁱ Pr ₂ guan-NMe ₂) ₃	H ₂ O			Ref. 174
Gd ₂ O ₃	Gd(ⁱ PrCp) ₃	H ₂ O			Ref. 217

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TABLE I. (Continued.)

Material	Reactant A	Reactant B	Reactant C	Further reactants	First Ref
Gd ₂ O ₃	Gd(ⁱ PrCp) ₃	O ₃			Ref. 218
Gd ₂ O ₃	Gd(ⁱ PrCp) ₂ (ⁱ Pr ₂ AMD)	O ₃			Ref. 175
GdAl _x O _y	Gd(Cp ⁱ Pr) ₃	AlMe ₃	O ₃		Ref. 219
GdAl _x O _y	Gd(Cp ⁱ Pr) ₃	AlMe ₃	H ₂ O		Ref. 219
GdCa _x Co _y O _z	Gd(thd) ₃	Ca(thd) ₂	Co(thd) ₂	O ₃	Ref. 220
GdCo _x O _y	Gd(thd) ₃	Co(thd) ₂	O ₃		Ref. 220
GdF ₃	Gd(thd) ₃	NH ₄ F			Ref. 221
GdGa _x Er _y O _z	Gd(thd) ₃	GaEt ₃	Er(thd) ₃	O ₃	Ref. 222
GdSc _x O _y	Gd(Cp ⁱ Pr) ₃	Sc(CpMe) ₃	H ₂ O		Ref. 49
GeO ₂	Ge(NMe ₂) ₂ (NH ⁱ Pr(CH ₂) ₂ NH ⁱ Pr)	O ₃			Ref. 223
GeO ₂	Ge(NMe ₂) ₂ (NH ⁱ Bu(CH ₂) ₂ NH ⁱ Bu)	O ₃			Ref. 223
GeO ₂	Ge(NMe ₂) ₄	O ₃			Ref. 224
GeO ₂	Ge(O ⁿ Bu) ₄	O ₃			Ref. 224
GeO ₂	Ge(NMe ₂) ₄	H ₂ O			Ref. 225
GeO ₂	Ge(thd)Cl	H ₂ O ₂			Ref. 226
GeO _x	Ge(NMe ₂) ₄	H ₂ O			Ref. 225
GeS	Ge(η ² -(N ⁱ Bu)CHMeCHMe(N ⁱ Bu)))	H ₂ S			Ref. 227
GeSb	GeCl ₂ ·(C ₄ H ₈ O ₂)	Sb(SiEt ₃) ₃			Ref. 48
GeSb _x Se _y Te _z	Ge(OEt) ₄	Sb(OEt) ₃	Se(SiMe ₃) ₂	Te(SiMe ₃) ₂	Ref. 228
GeSb _x Te _y	Ge(OMe) ₄	Sb(OEt) ₃	Te(SiMe ₃) ₂		Ref. 229
GeSb _x Te _y	Ge(OEt) ₄	Sb(OEt) ₃	Te(SiMe ₃) ₂		Ref. 230
GeSb _x Te _y	HGeCl ₃	Sb(OEt) ₃	Te(SiMe ₃) ₂		Ref. 231
GeSb _x Te _y	Ge(ⁱ Pr ₂ guan-NMe ₂)(NMe ₂)	Sb(OEt) ₃	Te(SiMe ₃) ₂ + NH ₃		Ref. 232
GeSe	HGeCl ₃	Se(SiMe ₃) ₂			Ref. 233
GeSe	Ge(N(SiMe ₃) ₂) ₂	Se(SiMe ₃) ₂	MeOH		Ref. 234
GeSe	Ge(N(SiMe ₃) ₂) ₂	Se(SiMe ₃) ₂			Ref. 234
GeS _x Se _y	HGeCl ₃	H ₂ S	Se(SiMe ₃) ₂		Ref. 235
GeTe	Ge(N(SiMe ₃) ₂) ₂	Te(SiMe ₃) ₂	MeOH		Ref. 236
GeTe	GeCl ₃ H	Te(SiMe ₃) ₂			Ref. 231
GeTe	Ge(ⁱ Pr ₂ guan-NMe ₂)(NMe ₂)	Te(SiMe ₃) ₂ + NH ₃			Ref. 237
GeTe ₂	Ge(OMe) ₄	Te(SiMe ₃) ₂			Ref. 229
GeTe ₂	Ge(OEt) ₄	Te(SiMe ₃) ₂			Ref. 230
GeTe ₂	GeCl ₄	Te(GeMe ₃) ₂			Ref. 238
GeTe _x	HGeCl ₃	Te(OEt) ₄	(SiEt ₃) ₂ Te		Ref. 239
HfAl _x C _y	HfCl ₄	AlMe ₃			Ref. 240
HfAl _x O _y	Hf(NEtMe) ₄	AlMe ₃	H ₂ O		Ref. 241
HfAl _x O _y	Hf(NEtMe) ₄	AlMe ₃	H ₂ O ₂		Ref. 242
HfAl _x Ti _y O _z	Hf(NMeEt) ₄	AlMe ₃	Ti(OEt) ₄	H ₂ O	Ref. 243
HfEr _x O _y	(MeCp) ₂ HfMe(OMe)	(ⁱ PrCp) ₃ Er	O ₃		Ref. 244
HfEr _x O _y	Hf(CpMe) ₂ (OMe)Me	Er(thd) ₃	O ₃		Ref. 245
HfF ₄	Hf(NEtMe) ₄	HF			Ref. 246
HfGd _x O _y	HfCl ₄	Gd(Cp ⁱ Pr) ₃	H ₂ O		Ref. 247
HfLa _x O _y	Hf(NEtMe) ₄	La(FMD) ₃	O ₃		Ref. 248
HfO ₂	Hf(Cp)(NMe ₂) ₃	O ₃			Ref. 249
HfO ₂	Hf(CpMe)(NMe ₂) ₃	O ₃			Ref. 249
HfO ₂	Hf(Cp)(NMe ₂) ₃	H ₂ O			Ref. 250
HfO ₂	Hf(NMe ₂) ₂ (ⁱ Pr ₂ guan-NMe ₂) ₂	H ₂ O			Ref. 251
HfO ₂	Hf(ⁱ Pr ₂ guan-NEtMe)(NEtMe) ₃	H ₂ O			Ref. 252
HfO ₂	Hf(BH ₄) ₄	H ₂ O			Ref. 253
HfO ₂	Hf(NMe ₂) ₄	O ₃	H ₂ O		Ref. 254

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TABLE I. (Continued.)

Material	Reactant A	Reactant B	Reactant C	Further reactants	First Ref
HfO ₂	HfCp(edpa) ₃	H ₂ O			Ref. 255
HfO ₂	[η ⁶ :η ¹ -(C ₅ H ₅ BNMe ₂)]Hf(NMe ₂) ₃	O ₂			Ref. 256
HfO ₂	[η ⁶ :η ¹ -(C ₅ H ₅ BNMe ₂)]Hf(NMe ₂) ₃	H ₂ O			Ref. 256
HfO ₂	Hf(CpMe ₂)(NMe ₂) ₃	O ₃			Ref. 257
HfO ₂	Hf(dmap) ₄	O ₃			Ref. 258
HfS ₂	HfCl ₄	H ₂ S			Ref. 259
HfS ₂	Hf(NMe ₂) ₄	H ₂ S			Ref. 260
HfSc _x O _y	Hf(NEt ₂) ₄	Sc(MeCp) ₃	H ₂ O		Ref. 261
HfTi _x O _y	Hf(NEtMe) ₄	TiCl ₄	H ₂ O		Ref. 262
HfY _x O _y	HfCl ₄	Y(CpMe) ₃	H ₂ O		Ref. 263
HfZn _x O _y	Hf(NEtMe) ₄	ZnEt ₂	H ₂ O		Ref. 264
HfZr _x O _y	Hf(NEt ₂) ₄	Zr(NEt ₂) ₄	H ₂ O		Ref. 265
HfZr _x O _y	Hf(NEtMe) ₄	Zr(NEtMe) ₄	H ₂ O		Ref. 266
HfZr _x O _y	Hf(NEtMe) ₄	Zr(NEtMe) ₄	O ₃		Ref. 266
HoF ₃	Ho(thd) ₃	NbF ₅			Ref. 123
HoYb _x Ti _y O _z	Ho(thd) ₃	Yb(thd) ₃	TiCl ₄	O ₃ /H ₂ O	Ref. 178
In ₂ O ₃	InCp	H ₂ O	O ₂		Ref. 267
In ₂ O ₃	In(ⁱ Pr ₂ guan-NEt ₂) ₃	H ₂ O			Ref. 268
In ₂ O ₃	In(N(SiMe ₃) ₂)Et ₂	H ₂ O			Ref. 269
In ₂ O ₃	InCp	H ₂ O	O ₂		Ref. 270
In ₂ O ₃	In(N(SiMe ₃) ₂)Et ₂	O ₃			Ref. 271
In ₂ O ₃	In(PrNMe ₂)Me ₂	H ₂ O			Ref. 270
In ₂ O ₃	In(PrNMe ₂)Me ₂	O ₃			Ref. 270
In ₂ O ₃	InEt ₃	O ₃			Ref. 272
In ₂ O ₃	In(dmamp) ₃	O ₃			Ref. 273
In ₂ O ₃	InMe ₂ (edpa)	H ₂ O			Ref. 274
In ₂ O ₃	In(ⁱ PrAMD) ₃	H ₂ O			Ref. 274
In ₂ O ₃	In(ⁱ PrFMD) ₃	H ₂ O			Ref. 275
In ₂ O ₃	InCp	H ₂ O ₂			Ref. 276
In ₂ O ₃	InMe ₃	O ₃			Ref. 277
In ₂ O ₃	In(EtCp)	H ₂ O	O ₃		Ref. 278
In ₂ O ₃	In(EtCp)	H ₂ O	O ₂		Ref. 278
In ₂ O ₃	In(dmamp) ₂ (O ⁱ Pr)	O ₃			Ref. 279
In ₂ O ₃	InMe ₃ (MeO(CH ₂) ₂ NH ⁱ Bu)	H ₂ O			Ref. 280
In ₂ O ₃	In(iPr ₂ triaz) ₃	H ₂ O			Ref. 281
In ₂ S ₃	In(ⁱ PrAMD) ₃	H ₂ S			Ref. 282
InAl _x O _y	In(PrNMe ₂)Me ₂	AlMe ₃	O ₃		Ref. 283
InAl _x Sn _y O _z	InMe ₃	AlMe ₃	Sn(NMe ₂) ₄	H ₂ O	Ref. 284
InGa _x As _y	InMe ₃	GaMe ₃	AsH ₃		Ref. 285
InGa _x N _y	InMe ₃	GaEt ₃	NH ₃ + Ar + N ₂		Ref. 286
InGa _x O _y	InN(SiMe ₃) ₂	GaMe ₃	H ₂ O ₂		Ref. 287
InGa _x O _y	In(PrNMe ₂)Me ₂	GaMe ₃	O ₃		Ref. 216
InGa _x Zn _y O _z	InMe ₃	GaEt ₃	ZnEt ₂	H ₂ O	Ref. 288
InGa _x Zn _y O _z	InCp	GaMe ₃	ZnEt ₂	O ₂	Ref. 289
InSe	InCl ₃	H ₂ Se			Ref. 290
InSn _x Ga _y O _z	InMe ₂ (edpa)	Sn(dmamp) ₂	GaMe ₃	O ₂	Ref. 291
InSn _x O _y	InMe ₃	Sn(NMe ₂) ₄	O ₃		Ref. 276
InTi _x O _y	In(PrNMe ₂)Me ₂	Ti(NMe ₂) ₄	H ₂ O	O ₃	Ref. 292
InZn _x O _y	InMe ₃	ZnEt ₂	H ₂ O		Ref. 293
InZn _x O _y	In(NMe ₂)Me ₂	ZnEt ₂	H ₂ O		Ref. 294
InZn _x O _y	In(PrNMe ₂)Me ₂	ZnEt ₂	O ₃		Ref. 216

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TABLE I. (Continued.)

Material	Reactant A	Reactant B	Reactant C	Further reactants	First Ref
Ir	Ir(CpMe)(CHD)	O ₂			Ref. 295
Ir	Ir(CpMe)(CHD)	O ₃	H ₂		Ref. 296
Ir	Ir(acac) ₃	O ₂	H ₂		Ref. 297
Ir	Ir(CpEt)(CHD)	O ₃			Ref. 298
Ir	Ir(CO) ₃ (^t Bu ₃ Cyp)	O ₂			Ref. 299
IrO ₂	Ir(CpMe)(CHD)	O ₃			Ref. 296
IrO ₂	Ir(CpEt)(CHD)	O ₃			Ref. 300
KAl _x O _y	K(O ^t Bu)	AlMe ₃	H ₂ O		Ref. 301
KNa _x Nb _y O _z	K(O ^t Bu)	Na(O ^t Bu)	Nb(OEt) ₅	H ₂ O	Ref. 302
KNa _x Ta _y O _z	K(O ^t Bu)	Na(O ^t Bu)	Ta(OEt) ₅	H ₂ O	Ref. 302
KNb _x O _y	K(O ^t Bu)	Nb(OEt) ₅	H ₂ O		Ref. 302
KNb _x O _y	K(O ^t Bu)	Nb(OEt) ₅	O ₃		Ref. 303
KNb _x Ta _y O _z	K(O ^t Bu)	Nb(OEt) ₅	Ta(OEt) ₅	H ₂ O	Ref. 302
KNb _x Ta _y O _z	K(O ^t Bu)	Nb(OEt) ₅	Ta(OEt) ₅	O ₃	Ref. 303
KO	K(O ^t Bu)	H ₂ O			Ref. 301
KP _x O _y N _z	K(O ^t Bu)	PO(NH) ₂ (OEt) ₂			Ref. 304
KTa _x O _y	K(O ^t Bu)	Ta(OEt) ₅	H ₂ O		Ref. 302
La ₂ O ₃	La(ⁱ PrFMD) ₃	O ₂			Ref. 305
La ₂ O ₃	La(ⁱ PrCp) ₂ (ⁱ PrAMD)	H ₂ O			Ref. 306
La ₂ O ₃	La(ⁱ PrCp) ₂ (ⁱ PrAMD)	O ₃			Ref. 306
La ₂ O ₃	La(ⁱ PrCp) ₂ (ⁱ PrAMD)	H ₂ O	O ₃		Ref. 306
La ₂ O ₃	La(ⁱ PrCp) ₂ (ⁱ PrAMD)	EtOH			Ref. 306
La ₂ O ₃	La(THD) ₃ (DMEA)	O ₃			Ref. 307
LaAl _x O _y	[LaAl(O ⁱ Pr) ₆ (ⁱ PrOH)] ₂	H ₂ O			Ref. 308
LaAl _x O _y	La(Cp ⁱ Pr) ₃	AlMe ₃	H ₂ O		Ref. 309
LaAl _x O _y	La(thd) ₃	AlMe ₃	O ₃		Ref. 310
LaCa _x P _y O _z	La(thd) ₃	Ca(thd) ₂	PO(OMe) ₃	O ₃ / O ₃ +H ₂ O	Ref. 311
LaCu _x O _y	La(thd) ₃	Cu(acac) ₂	O ₃		Ref. 312
LaF ₃	La(thd) ₃	O ₃	Hhfac	O ₃	Ref. 88
LaNi _x O _y	La(thd) ₃	Ni(acac) ₂	O ₃		Ref. 313
LaP _x O _y	La(thd) ₃	PO(OMe) ₃	O ₃	H ₂ O + O ₃	Ref. 311
LaSc _x O _y	La(thd) ₃	Sc(thd) ₃	O ₃		Ref. 176
LaSr _x Co _y O _z	La(thd) ₃	Sr(thd) ₂	Co(acac) ₃	O ₃	Ref. 314
LaSr _x Cu _y O _z	La(thd) ₃	Sr(thd) ₂	Cu(acac) ₂	O ₃	Ref. 315
LaSr _x Ti _y O _z	La(ⁱ PrFMD) ₃	Sr(Cp ⁱ Pr) ₂	Ti(O ⁱ Pr) ₄	H ₂ O	Ref. 316
LaTi _x O _y	La(thd) ₃	TiCl ₄	O ₃	H ₂ O	Ref. 317
LaW _x O _y	W(N ^t Bu) ₂ (NMe ₂) ₂	H ₂ O	La(thd) ₃	O ₃	Ref. 318
LaYb _x Ti _y O _z	La(thd) ₃	Yb(thd) ₃	TiCl ₄	O ₃ /H ₂ O	Ref. 178
Li ₂ CO ₃	Li(OSiMe ₃)	H ₂ O	CO ₂		Ref. 319
Li ₂ CO ₃	Li(O ^t Bu)	H ₂ O	CO ₂		Ref. 320
Li ₂ O	Li(O ^t Bu)	H ₂ O			Ref. 317
Li ₂ S	Li(O ^t Bu)	H ₂ S			Ref. 321
Li ₃ N	Li(N(SiMe ₃) ₂)	NH ₃			Ref. 322
Li ₄ Ti ₅ O ₁₂	Li(N(SiMe ₃) ₂)	Ti(NMe ₂) ₄	H ₂ O		Ref. 323
LiAl _x F _y	Li(O ^t Bu)	AlCl ₃	TiF ₄		Ref. 324
LiAl _x O _y	Li(O ^t Bu)	AlMe ₃	H ₂ O	O ₃ H ₂ O	Ref. 325
LiAl _x O _y	Li(OSiMe ₃)	H ₂ O	AlMe ₃	O ₃	Ref. 319
LiAl _x Si _y O _z	Li(O ^t Bu)	AlMe ₃	Si(OEt) ₄	H ₂ O	Ref. 326
LiAl _x S _y	Li(O ^t Bu)	Al(NMe ₂) ₃	H ₂ S		Ref. 327
LiB _x C _y O _z	Li(O ^t Bu)	B(O ⁱ Pr) ₃	O ₃		Ref. 328
LiCo _x O _y	Li(thd)	Co(thd) ₂	O ₃		Ref. 329

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TABLE I. (Continued.)

Material	Reactant A	Reactant B	Reactant C	Further reactants	First Ref
LiC _x O _y	Li(N(SiMe ₃) ₂)	CO ₂	H ₂ O		Ref. 322
LiC _x O _y	Li(O ^t Bu)	CO ₂	H ₂ O		Ref. 330
LiF	Li(thd)	TiF ₄	Mg(thd) ₂		Ref. 331
LiF	Li(thd)	TiF ₄			Ref. 332
LiF	Li(N(SiMe ₃) ₂)	HF-pyridine			Ref. 333
LiF	Li(O ^t Bu)	TiF ₄			Ref. 324
LiF	Li(O ^t Bu)	HF			Ref. 334
LiF	Li(N(SiMe ₃) ₂)	HF			Ref. 335
LiF	Li(O ^t Bu)	NH ₄ F			Ref. 336
LiF-CF _x	Li(O ^t Bu)	Hfac			Ref. 337
LiFe _x P _y O _z	Li(O ^t Bu)	FeCp ₂	PO(OMe) ₃	H ₂ O/O ₃	Ref. 338
LiLa _x Ti _y O _z	Li(O ^t Bu)	La(thd) ₃	TiCl ₄	H ₂ O/O ₃	Ref. 317
LiLa _x Zr _y O _z	Li(O ^t Bu)	La(ⁱ PrFMD) ₃	Zr(NMe ₂) ₄	H ₂ O/O ₃	Ref. 339
LiMn _x O _y	Li(thd)	Mn(thd) ₃	O ₃		Ref. 340
LiMn _x O _y	Mn(thd) ₃	O ₃	Li(O ^t Bu)	H ₂ O	Ref. 341
LiNb _x O _y	Li(N(SiMe ₃) ₂)	Nb(OEt) ₅	H ₂ O		Ref. 342
LiNb _x O _y	Li(O ^t Bu)	Nb(OEt) ₅	H ₂ O		Ref. 343
LiOH	Li(O ^t Bu)	H ₂ O			Ref. 317
LiP _x O _y	Li(N(SiMe ₃) ₂)	PO(OMe) ₃			Ref. 344
LiP _x O _y	Li(O ^t Bu)	PO(OMe) ₃			Ref. 344
LiP _x O _y	Li(O ^t Bu)	PO(OMe) ₃	H ₂ O		Ref. 345
LiP _x O _y	Li(O ^t Bu)	P(OMe) ₃	H ₂ O		Ref. 47
LiP _x O _y	Li(N(SiMe ₃) ₂)	PO(OMe) ₃	AlMe ₃		Ref. 346
LiP _x O _y N _z	Li(N(SiMe ₃) ₂)	PO(NH ₂)(OEt) ₂			Ref. 347
LiP _x O _y N _z	Li(O ^t Bu)	P(NMe ₂) ₃	O ₂	NH ₃	Ref. 348
LiP _x O _y N _z	Li(O ^t Bu)	PO(NH ₂)(OEt) ₂			Ref. 349
LiSi _x O _y	Li(N(SiMe ₃) ₂)	O ₃			Ref. 350
LiSi _x O _y	Li(OSiMe ₃)	O ₃	H ₂ O		Ref. 319
LiSi _x O _y	Li(O ^t Bu)	Si(OEt) ₄	H ₂ O		Ref. 351
LiTa _x O _y	Li(O ^t Bu)	Ta(OEt) ₅	H ₂ O		Ref. 352
LiTi _x O _y	Li(O ^t Bu)	Ti(O ⁱ Pr) ₄	H ₂ O		Ref. 353
LiV _x O _y	Li(thd)	VO(thd) ₂	O ₃		Ref. 340
LiZr _x O _y	Li(O ^t Bu)	Zr(NMe ₂) ₄	H ₂ O		Ref. 354
Lu ₂ O ₃	Lu(ⁱ PrCp) ₃	H ₂ O			Ref. 355
Lu ₂ O ₃	Lu(ⁱ PrCp) ₃	O ₃			Ref. 355
Lu ₂ O ₃	Lu(thd) ₃	O ₃			Ref. 356
LuAl _x O _y	Lu(ⁱ PrCp) ₃	Al(NEt ₂) ₃	O ₃		Ref. 355
LuAl _x O _y	Lu(ⁱ PrCp) ₃	Al(NEt ₂) ₃	H ₂ O		Ref. 355
LuAl _x O _y	Lu(thd) ₃	AlMe ₃	O ₃		Ref. 356
LuMn _x O _y	Lu(thd) ₃	Mn(thd) ₃	O ₃		Ref. 357
LuSc _x O _y	Lu(thd) ₃	Sc(thd) ₃	O ₃		Ref. 176
MgCa _x O _y	Mg(^s BuAMD) ₂	Ca(ⁱ PrAMD) ₂	H ₂ O		Ref. 358
MgF ₂	Mg(thd) ₂	O ₃	Hhfac	O ₃	Ref. 88
MgF ₂	Mg(EtCp) ₂	HF			Ref. 359
MgO	MgCp ₂	O ₃			Ref. 360
Mn ₃ (BO ₃) ₂	MnTp ₂	O ₃			Ref. 121
Mn ₅ O ₈	Mn(EtCp) ₂	O ₃			Ref. 361
MnCo _x O _y	Mn(thd) ₃	Co(thd) ₃	O ₃		Ref. 362
MnF ₂	Mn(EtCp) ₂	HF			Ref. 246
MnO _x	Mn ₂ (CO) ₁₀	O ₃			Ref. 363
MnO _x	Mn(^t BuAMD) ₂	H ₂ O			Ref. 364

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TABLE I. (Continued.)

Material	Reactant A	Reactant B	Reactant C	Further reactants	First Ref
MnS	Mn(EtCp) ₂	H ₂ S			Ref. 365
Mo	MoF ₆	Si ₂ H ₆			Ref. 366
Mo	MoO ₂ Cl ₂	H ₂			Ref. 367
Mo	MoO ₂ Cl ₂	H (hot wire)			Ref. 368
MoC _x	Mo(EtBen) ₂	H ₂			Ref. 369
MoN _x	Mo(CO) ₆	NH ₃			Ref. 370
MoN _x	MoCl ₄ (THD)(THF)	H ₂	NH ₃		Ref. 371
MoO ₃	Mo(CO) ₆	O ₃			Ref. 372
MoO ₃	Mo(N ^t Bu) ₂ (NMe ₂) ₂	O ₃			Ref. 373
MoON	MoO ₂ (^t BuAMD) ₂	O ₃			Ref. 374
MoO _x	Mo(EtBen) ₂	H ₂ O			Ref. 375
MoO _x	Mo(SiMe ₃ Cp)(CO) ₂ (η ³ -2-MeAllyl)	O ₃			Ref. 376
MoO _x	MoO ₂ (^t BuAMD) ₂	O ₃			Ref. 374
MoO _x	MoO ₂ (ⁱ PrAMD) ₂	O ₃			Ref. 377
MoO _x	Mo(NMe ₂) ₄	O ₃			Ref. 378
MoO _x	Mo(NMe ₂) ₄	H ₂ O			Ref. 378
MoO _x	Mo(N ^t Bu) ₂ (ⁱ PrAMD) ₂	O ₃			Ref. 379
MoO _x	MoO ₂ (thd) ₂	O ₃			Ref. 380
MoO _x	MoCl ₄ O	O ₃ + H ₂ O			Ref. 381
MoO _x	MoOCl ₄	H ₂ O			Ref. 382
MoO _x	Mo(ⁱ PrCp) ₂ H ₂	O ₃			Ref. 383
MoO _x	Mo(N ^t Bu) ₂ (NMe ₂) ₂	H ₂ O ₂			Ref. 384
MoO _x N _γ	Mo(CO) ₆	O ₃	NH ₃		Ref. 385
MoP	MoCl ₅	PH ₃			Ref. 386
MoS ₂	Mo(CO) ₆	S ₂ Me ₂			Ref. 387
MoS ₂	Mo(CO) ₆	H ₂ S			Ref. 388
MoS ₂	MoCl ₅	H ₂ S			Ref. 389
MoS ₂	Mo(CO) ₆	EtSSEt	S ₂ Et ₂		Ref. 390
MoS ₂	Mo(CO) ₆	S(SiMe ₃) ₂			Ref. 391
MoS ₂	Mo(NMe ₂) ₄	HS(CH ₂) ₂ SH			Ref. 392
MoS ₂	Mo(NMe ₂) ₄	H ₂ S			Ref. 393
MoS ₂	Mo(thd) ₃	H ₂ S			Ref. 394
MoS ₂	MoF ₆	H ₂ S			Ref. 395
MoS ₂	Mo(NMe ₂) ₂ (NtBu) ₂	PrSH			Ref. 396
MoS ₂	Mo(NMe ₂) ₂ (NtBu) ₂	H ₂ S			Ref. 397
MoS ₂	MoCl ₅	S(SiMe ₃) ₂			Ref. 398
MoS ₂	(HCHT)Mo(CO) ₃	H ₂ S			Ref. 399
MoSe ₂	Mo(CO) ₆	Se(SiMe ₃) ₂			Ref. 400
MoSe ₂	MoCl ₅	Se(SiMe ₃) ₂			Ref. 400
MoSe ₂	MoCl ₅	Se(Si ₂ Me ₄) ₂ Se			Ref. 401
MoSe ₂	MoCl ₅	Se(SiMe ₂ (CH ₂) ₂ SiMe ₂)			Ref. 401
MoSe ₂	MoCl ₅	Se(SiEt ₃) ₂			Ref. 402
MoSe ₂	MoCl ₅	Se(Si ⁱ Pr ₃) ₂			Ref. 402
MoSe ₂	MoCl ₅	Se(SnMe ₃) ₂			Ref. 402
MoSe ₂	MoCl ₅	Se ₂ (Si(ⁱ Pr) ₂) ₂			Ref. 403
MoSe _x O _γ	Mo(CO) ₆	(Me ₃ Si) ₂ Se			Ref. 404
MoTe ₂	MoCl ₅	Me ₈ Si ₄ Te ₂			Ref. 405
MoTi _x O _γ	MoOCl ₄	TiCl ₄	H ₂ O		Ref. 384
NaAl _x O _γ	Na(OSiMe ₃)	AlMe ₃	H ₂ O		Ref. 301
NaAl _x O _γ	Na(O ^t Bu)	AlMe ₃	O ₃		Ref. 301
NaAl _x O _γ	Na(O ^t Bu)	AlMe ₃	H ₂ O		Ref. 301

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TABLE I. (Continued.)

Material	Reactant A	Reactant B	Reactant C	Further reactants	First Ref
NaF	Na(O ^t Bu)	HF			Ref. 406
NaMn _x O _y	Na(O ^t Bu)	H ₂ O	Mn(thd) ₃	O ₃	Ref. 407
NaNb _x O _y	Na(O ^t Bu)	Nb(OEt) ₅	H ₂ O		Ref. 302
NaNb _x Ta _y O _z	Na(O ^t Bu)	Nb(OEt) ₅	Ta(OEt) ₅	H ₂ O	Ref. 302
NaO	Na(O ^t Bu)	H ₂ O			Ref. 301
NaP _x O _y N _z	Na(O ^t Bu)	PO(NH ₂)(OEt) ₂			Ref. 408
NaTa _x O _y	Na(O ^t Bu)	Ta(OEt) ₅	H ₂ O		Ref. 302
NaTi _x O _y	Na(O ^t Bu)	Ti(O ⁱ Pr) ₄	H ₂ O		Ref. 409
Na _x CoO ₂	Na(thd)	Co(acac) ₃	O ₃		Ref. 410
Nb ₂ O ₅	Nb(N ^t Bu)(NEt ₂) ₃	H ₂ O			Ref. 411
Nb ₂ O ₅	Nb(N ^t Bu)(NEt ₂) ₃	O ₃			Ref. 411
Nb ₂ O ₅	Nb(N ^t Bu)(NEtMe) ₃	O ₃			Ref. 411
Nb ₂ O ₅	Nb(N ^t Bu)(NEtMe) ₃	H ₂ O			Ref. 411
Nb ₂ O ₅	Nb(OEt) ₅	Photons			Ref. 412
Nb ₂ O ₅	Nb(OEt) ₅	O ₃			Ref. 413
Nb ₂ O ₅	Nb(N ^t Bu)(NEt ₂) ₂ (Cp)	O ₃			Ref. 414
NbAl _x O _y	Nb(OEt) ₅	AlMe ₃	H ₂ O		Ref. 415
NbC _x	NbCl ₅	AlMe ₃			Ref. 416
NbC _x	NbF ₅	AlMe ₃			Ref. 416
NbSi	NbF ₅	Si ₂ H ₆			Ref. 417
NbSi _x O _y	Nb(OEt) ₅	Si ₂ (NH ₂) ₆	O ₃		Ref. 413
NbSn _x O _y	Nb(N ^t Bu)(NEt ₂) ₃	Sn(NMe ₂) ₄	O ₃		Ref. 418
NbS _x	Nb(N ^t Bu)(NEt ₂) ₃	H ₂ S			Ref. 419
NbTa _x O _y	NbF ₅	TaF ₅	H ₂ O	O ₃	Ref. 420
NbTi _x N _y	NbCl ₅	TiCl ₄	NH ₃		Ref. 421
NdAl _x O _y	[NdAl(O ⁱ Pr) ₆ (ⁱ PrOH)] ₂	H ₂ O			Ref. 53
NdNi _x O _y	Nd(^t PrCp) ₃	Ni(^t BuAMD) ₂	O ₃		Ref. 422
NdTi _x O _y	Nd(thd) ₃	TiCl ₄	O ₃	H ₂ O	Ref. 178
NdYb _x Ti _y O _z	Nd(thd) ₃	Yb(thd) ₃	TiCl ₄	O ₃ / H ₂ O	Ref. 178
Ni	Ni(dmamb) ₂	NH ₃			Ref. 423
Ni	Ni(Me ⁱ PrCOCN ^t Bu) ₂	BH ₃ (NHMe ₂)			Ref. 105
Ni	Ni(acac) ₂	CH ₃ OH			Ref. 424
Ni	Ni(^t Bu ₂ DAD) ₂	^t BuNH ₂			Ref. 425
Ni	Ni(acac) ₂ (pyr) ₂	H ₂ NNH ₂			Ref. 426
Ni	Ni(acac) ₂ (TMEDA)	H ₂ NNH ₂			Ref. 426
Ni	Ni(Chex)(Cp)	NH ₃			Ref. 427
Ni	NiCl ₂ (PEt ₃) ₂	(Me ₃ Ge) ₂ DHP			Ref. 428
Ni ₂ Ge	NiCl ₂ (TMPDA)	GeH(ⁿ Bu) ₃			Ref. 429
Ni ₃ N	NiCl ₂ (TMPDA)	^t BuNHNH ₂			Ref. 430
Ni ₃ Sn ₂	NiCl ₂ (TMPDA)	SnH ₂ Sn			Ref. 118
NiCo _x O _y	Ni(thd) ₂	Co(thd) ₂	O ₃		Ref. 431
NiC _x	Ni(acac) ₂	EtOH			Ref. 424
NiC _x	Ni(acac) ₂	ⁿ PrOH			Ref. 424
NiC _x	Ni(acac) ₂ (TMEDA)	HCOOH			Ref. 432
NiF ₂	Ni(thd) ₂	NbF ₅			Ref. 123
NiFe _x O _y	NiCp ₂	FeCp ₂	O ₃		Ref. 124
NiFe _x O _y	Ni(acac) ₂	Fe(Cp) ₂	O ₃		Ref. 433
NiFe _x O _y	NiCp ₂	Fe(Cp)(^t BuAMD)	O ₃		Ref. 434
NiO	Ni(dmamb) ₂	O ₃			Ref. 435
NiO	Ni(MeCp) ₂	H ₂ O			Ref. 436
NiO	Ni(^t BuAMD) ₂	H ₂ O			Ref. 437

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TABLE I. (Continued.)

Material	Reactant A	Reactant B	Reactant C	Further reactants	First Ref
NiO	Ni(acac) ₂ (TMEDA)	O ₃			Ref. 438
NiO	Ni(^t Bu ₂ DAD) ₂	O ₃			Ref. 439
NiO	Ni(dmamb) ₂	H ₂ O ₂			Ref. 440
NiS	Ni(thd) ₂	H ₂ S			Ref. 441
NiS	Ni(acac) ₂	H ₂ S			Ref. 442
NiS _x	Ni(^t BuAMD) ₂	H ₂ S			Ref. 443
NiS _x	Ni(dmamb) ₂	H ₂ S			Ref. 444
NiS _x	Ni(^t BuAMD) ₂	^t BuSS ^t Bu			Ref. 445
NiTi _x O _y	Ni(acac) ₂	Ti(O ⁱ Pr) ₄	O ₃	H ₂ O	Ref. 446
NiZn _x	Ni(thd) ₂	ZnEt ₂			Ref. 171
Os	OsCp ₂	O ₂			Ref. 447
PbBr ₂	Pb(N(SiMe ₃) ₂) ₂	SnBr ₄			Ref. 448
PbBr ₂	Pb(N(SiMe ₃) ₂) ₂	TiBr ₄			Ref. 448
PbBr _x Cl _y	Pb(N(SiMe ₃) ₂) ₂	TiBr ₄	GaCl ₃		Ref. 448
PbBr _x I _y	Pb(N(SiMe ₃) ₂) ₂	TiBr ₄	SnI ₄		Ref. 448
PbCl ₂	[Pb(gem)] ₂	SnCl ₄			Ref. 448
PbCl ₂	Pb(N(SiMe ₃) ₂) ₂	SnCl ₄			Ref. 448
PbCl ₂	Pb(N(SiMe ₃) ₂) ₂	TiCl ₄			Ref. 448
PbCl ₂	Pb(N(SiMe ₃) ₂) ₂	GaCl ₃			Ref. 448
PbHf _x O _y	Pb(dmamp) ₂	Hf(NEtMe) ₄	H ₂ O	O ₃	Ref. 449
PbHf _x Ti _y O _z	Pb(dmamp) ₂	Hf(NEtMe) ₄	Ti(NMe ₂) ₄	H ₂ O + O ₃	Ref. 449
PbI ₂	Pb(N(SiMe ₃) ₂) ₂	SnI ₄			Ref. 450
PbI ₂	Pb(thd) ₂	HI			Ref. 451
PbI _x Cl _y	Pb(N(SiMe ₃) ₂) ₂	SnI ₄	GaCl ₃		Ref. 448
PbS	Pb(dbda)	H ₂ S			Ref. 452
PbS	Pb(N(SiMe ₃) ₂) ₂	H ₂ S			Ref. 452
PbSe _x Te _y	Pb(thd) ₂	Se(SiEt ₃) ₂	Te(SiEt ₃) ₂		Ref. 453
PbTi _x O _y	Pb(dmamp) ₂	Ti(NMe ₂) ₄	H ₂ O	O ₃	Ref. 454
PbZr _x Ti _y O _z	Pb(thd) ₂	ZrCl ₄	Ti (O ⁱ Pr) ₂ (thd) ₂	H ₂ O	Ref. 455
Pd	Pd(hfac) ₂	Formalin			Ref. 456
Pd	Pd(thd) ₂	O ₃	H ₂		Ref. 457
Pd	Pd(hfac) ₂	O ₃			Ref. 458
PdO	Pd(thd) ₂	O ₃			Ref. 457
PrAl _x O _y	Pr(Cp ⁱ Pr) ₃	AlMe ₃	H ₂ O		Ref. 459
PrO _x	Pr(EtCp) ₃	H ₂ O			Ref. 460
PrO _x	Pr(ⁱ PrCp) ₂ (ⁱ Pr ₂ AMD)	H ₂ O			Ref. 175
PrO _x	Pr(thd) ₃	O ₃			Ref. 461
PrTi _x O _y	Pr(thd) ₃	TiCl ₄	O ₃	H ₂ O	Ref. 178
PrYb _x Ti _y O _z	Pr(thd) ₃	Yb(thd) ₃	TiCl ₄	O ₃ / H ₂ O	Ref. 178
Pt	Pt(acac) ₂	O ₃	H ₂		Ref. 457
Pt	Pt(CpMe)Me ₃	O ₃			Ref. 462
Pt	PtMe ₂ (η ⁴ -cyclohexa-1,5-diene)	O ₂			Ref. 463
Pt	PtMe ₂ (dmba)	O ₂			Ref. 464
PtNi _x	(MeCp)PtMe ₃	O ₂	Ni(dmamb) ₂	H ₂	Ref. 465
RbNb _x O _y	Rb(O ^t Bu)	Nb(OEt) ₅	H ₂ O		Ref. 466
RbTi _x O _y	Rb(O ^t Bu)	Ti(O ⁱ Pr) ₄	H ₂ O		Ref. 466
RbTi _x O _y	Rb(O ^t Bu)	Ti(O ⁱ Pr) ₄	H ₂ O		Ref. 466
Re	ReCl ₅	NH ₃			Ref. 467
ReAl _x O _y	ReO ₃ Me	AlMe ₃			Ref. 468
ReN _x	ReCl ₅	NH ₃			Ref. 467

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TABLE I. (Continued.)

Material	Reactant A	Reactant B	Reactant C	Further reactants	First Ref
ReN _x	ReO ₃ Me	Me ₂ NNH ₂			Ref. 469
ReS ₂	ReCl ₅	H ₂ S			Ref. 470
Rh	Rh(acac) ₃	O ₃	H ₂		Ref. 457
Rh	Rh(acac) ₃	O ₃			Ref. 471
Ru	Ru(CpEt)(CpMe)	O ₂			Ref. 472
Ru	RuCp(CpCH(Me)(NMe ₂))	O ₂			Ref. 473
Ru	Ru(EtCp)(pyr)	O ₂			Ref. 474
Ru	Ru(ⁱ PrMeBen)(1,3-cyclohexadiene)	O ₂			Ref. 475
Ru	Ru(DMPD) ₂	O ₂			Ref. 476
Ru	Ru(Etben)(EtCHD)	O ₂			Ref. 477
Ru	Ru(EtCp) ₂	O ₃			Ref. 478
Ru	Ru(Me ₂ pyr) ₂	O ₂			Ref. 479
Ru	RuCp(CO) ₂ Et	O ₂			Ref. 480
Ru	Ru(EtBen)(1,3-cyclohexadiene)	O ₂			Ref. 481
Ru	Ru(CpEt)(dmopd)	O ₂			Ref. 482
Ru	Ru(ⁱ PrMeBen)(1,5-hexadiene)	O ₂			Ref. 483
Ru	Ru(CpEt) ₂	O ₂	H ₂		Ref. 484
Ru	Ru(EtBen)(1,3-butadiene)	O ₂			Ref. 485
Ru	Ru(CO) ₂ (mhdh) ₂	H ₂			Ref. 486
Ru	Ru(CO) ₂ (mhdh) ₂	NH ₃			Ref. 487
Ru	Ru(DMBD)(CO) ₃	O ₂			Ref. 488
Ru	Ru(CO) ₂ (mhdh) ₂	H ₂ O			Ref. 489
Ru	Ru(DMBD)(CO) ₃	H ₂ O			Ref. 490
Ru	Ru(DMBD)(CO) ₃	Me ₂ NNH ₂			Ref. 491
Ru	Ru(DMBD)(CO) ₃	N ₂ H ₄			Ref. 491
Ru	Ru(DMBD)(CO) ₃	^t BuNH ₂			Ref. 491
Ru	Ru(EtBen)(1,3-cyclohexadiene)	O ₂	H ₂		Ref. 492
Ru	Ru(CpEt)(DMOPD)	O ₂	NH ₃		Ref. 493
Ru	Ru(DMBD)(CO) ₃	HCOOH			Ref. 494
Ru	Ru(chd) ₂	O ₂			Ref. 495
Ru	Ru(TMM)(CO) ₃	O ₂			Ref. 496
Ru	Ru ₂ {μ ₂ -η ³ -N(^t Bu)-C(H)-C(ⁱ Pr)}(CO) ₆	H ₂ O			Ref. 497
Ru	Ru(CO) ₂ (mhdh) ₂	O ₂			Ref. 498
Ru	Ru(η ⁵ -C ₇ H ₇ O)(η ⁵ -C ₇ H ₉)	O ₂			Ref. 499
RuAl _x O _y	Ru(ⁱ PrMeBen)(1,3-cyclohexadiene)	AlMe ₃	O ₂	H ₂ O	Ref. 500
RuAl _x O _y	RuO ₄	AlMe ₃			Ref. 501
RuO ₂	Ru(thd) ₂ (cod)	O ₂			Ref. 502
RuO ₂	Ru(DMPD) ₂	O ₂			Ref. 476
RuO ₂	Ru(EtBen)(1,3-cyclohexadiene)	O ₂			Ref. 503
RuO ₂	Ru(ⁱ PrMeBen)(1,5-hexadiene)	O ₂			Ref. 483
RuO ₂	Ru(CpEt)(dmopd)	O ₂			Ref. 493
RuO ₂	Ru(TMM)(CO) ₃	O ₂			Ref. 504
RuO ₂	RuO ₄	CH ₃ OH			Ref. 505
RuO ₂	RuO ₄	CH ₃ CH ₂ OH			Ref. 505
RuO ₂	RuO ₄	CH ₃ (CH ₂) ₂ OH			Ref. 505
RuO ₂	RuO ₄	CH ₃ CHOHCH ₃			Ref. 505
RuPt _x	Ru(CO) ₂ (mhdh) ₂	Pt(DDAP)	O ₂		Ref. 506
RuPt _x O _y	RuO ₄	(MeCp)PtMe ₃			Ref. 501
Sb	SbCl ₃	(Et ₃ Si) ₃ Sb			Ref. 48
Sb	SbCl ₃	(Me ₃ Si) ₃ Sb			Ref. 507

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TABLE I. (Continued.)

Material	Reactant A	Reactant B	Reactant C	Further reactants	First Ref
Sb	Sb(NMe ₂) ₃	(SiMe ₃) ₃ Sb			Ref. 508
Sb	Sb(OEt) ₃	(SiMe ₃) ₃ Sb			Ref. 508
Sb ₂ Se ₃	SbCl ₃	H ₂ Se			Ref. 290
Sb ₂ Se ₃	Sb(OEt) ₃	Se(SiMe ₃) ₂			Ref. 228
Sb ₂ Se ₃	Sb(NMe ₂) ₃	Se(S ₂ CNMe ₂) ₄			Ref. 509
Sb ₂ Te ₃	SbCl ₃	Te(SiMe ₃) ₂			Ref. 510
Sb ₂ Te ₃	Sb(OEt) ₃	Te(SiMe ₃) ₂			Ref. 229
Sb ₂ Te ₃	Sb(NMe ₂) ₃	Te(GeMe ₃) ₂			Ref. 511
Sb ₂ Te ₃	Sb(OEt) ₃	Te(SiMe ₃) ₂ + NH ₃			Ref. 232
SbO _x	Sb(NMe ₂) ₃	H ₂ O ₂			Ref. 512
SbO _x	SbCl ₅	H ₂ O ₂			Ref. 512
SbO _x	SbCl ₅	Sb(OEt) ₃			Ref. 513
SbSe _x Te _y	Sb(NMe ₂) ₃	Se(SiMe ₃) ₂	Te ^t Bu ₂		Ref. 514
SbSe _x Te _y	Sb(OEt) ₃	Se(SiMe ₃) ₂	Te(SiMe ₃) ₂		Ref. 228
Sc ₂ O ₃	Sc(MeCp) ₃	H ₂ O			Ref. 49
Sc ₂ O ₃	Sc(MeCp) ₂ (Me ₂ pz)	O ₃			Ref. 515
Si	Si ₂ H ₆	Electrons			Ref. 516
Si ₃ N ₄	Si ₃ Cl ₈	NH ₃			Ref. 517
Si ₃ N ₄	SiL ₄	NH ₃			Ref. 518
SiO ₂	Si(OMe) ₄	H ₂ O + NH ₃ catalyst			Ref. 519
SiO ₂	SiH ₂ (NEt ₂) ₂	O ₃			Ref. 520
SiO ₂	SiH ₂ (NH ^t Bu) ₂	O ₃			Ref. 521
SiO ₂	SiH ₃ (N ^s Bu ₂)	O ₃			Ref. 521
SiO ₂	SiH ₃ (N ⁱ Pr ₂)	O ₃			Ref. 522
SiO ₂	(N ⁱ Pr ₂)H ₂ Si-SiH ₂ (N ⁱ Pr ₂)	O ₃			Ref. 523
SiO ₂	SiCl ₄	NH ₃	H ₂ O		Ref. 524
SiO ₂	(Me ₂ N) ₃ Si-SiH ₃	O ₃			Ref. 525
SiO ₂	N(SiMe ₃)(SiHMeNMe ₂) ₂	O ₃			Ref. 526
SiO ₂	Si ₂ H ₆	Electrons	O ₃		Ref. 527
SiO ₂	Si ₂ H ₆	Electrons	H ₂ O		Ref. 527
SiTi _x O _y	Si(O ⁿ Bu) ₄	Ti(NEt ₂) ₄			Ref. 522
SiTi _x O _y	SiH(NMe ₂) ₃	Ti(NMe ₂) ₄	H ₂ O		Ref. 528
SmMn _x O _y	Sm(thd) ₃	Mn(thd) ₃	O ₃		Ref. 357
SmTi _x O _y	Sm(thd) ₃	TiCl ₄	O ₃	H ₂ O	Ref. 178
SmYb _x Ti _y O _z	Sm(thd) ₃	Yb(thd) ₃	TiCl ₄	O ₃ / H ₂ O	Ref. 178
Sn	SnCl ₄	(SiMe ₃) ₂ DHP			Ref. 529
SnAl _x O _y	Sn(η ² -((N ^t Bu)CHMeCHMe(N ^t Bu))) ₂	AlMe ₃	H ₂ O ₂	H ₂ O	Ref. 530
SnAl _x O _y	Sn(NMe ₂) ₄	AlMe ₃	O ₃		Ref. 531
SnGa _x O _y	Sn(NMe ₂) ₄	Ga(NMe ₂) ₃	H ₂ O		Ref. 532
SnGa _x O _y	Sn(dmamp) ₂	GaMe ₃	O ₂		Ref. 291
SnGe _x O _y	Sn(NMe ₂) ₄	Ge(OMe) ₄	H ₂ O ₂		Ref. 533
SnGe _x O _y	Sn(NMe ₂) ₄	Ge(NMe ₂) ₄	H ₂ O		Ref. 225
SnI ₂	Sn(N(SiMe ₃) ₂) ₂	SnI ₄			Ref. 138
SnN _x	Sn(NMe ₂) ₄	NH ₃			Ref. 534
SnO	Sn(dmamp) ₂	H ₂ O			Ref. 535
SnO	Sn(N(SiMe ₃) ₂) ₂	H ₂ O			Ref. 536
SnO	Sn(η ² -((N ^t Bu)CMe ₂ CH ₂ (N ^t Bu)))	H ₂ O			Ref. 537
SnO	Sn(edpa) ₂	H ₂ O			Ref. 538
SnO	Sn(dmamb) ₂	H ₂ O			Ref. 539
SnO	Sn(ⁱ Pr ₂ FMD) ₂	H ₂ O			Ref. 540

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TABLE I. (Continued.)

Material	Reactant A	Reactant B	Reactant C	Further reactants	First Ref
SnO	Sn(O ^t Amyl) ₂	H ₂ O			Ref. 541
SnO ₂	Sn(O ^t Bu) ₄	CH ₃ COOH			Ref. 542
SnO ₂	Sn(η ² -((N ^t Bu)CHMeCHMe(N ^t Bu))) ₂	H ₂ O ₂			Ref. 543
SnO ₂	Sn(η ² -((N ^t Bu)CHMeCHMe(N ^t Bu))) ₂	NO			Ref. 544
SnO ₂	Sn(acac) ₂	O ₃			Ref. 545
SnO ₂	Sn(dmamp) ₂	O ₃			Ref. 546
SnO ₂	SnEt ₄	O ₃			Ref. 547
SnO ₂	SnEt ₄	H ₂ O ₂			Ref. 548
SnO ₂	Sn ⁿ Bu ₃ (OEt)	O ₃			Ref. 549
SnO ₂	Sn(η ² -((N ^t Bu)CMe ₂ CH ₂ (N ^t Bu)))	O ₃			Ref. 537
SnO ₂	Sn(acac) ₂	H ₂ O			Ref. 550
SnO ₂	SnI ₄	O ₃			Ref. 551
SnO _x P _γ	Sn(NMe ₂) ₄	H ₂ O	PO(OMe) ₃	O ₃	Ref. 552
SnO _x P _γ	Sn(NMe ₂) ₄	PO(OMe) ₃	O ₃		Ref. 552
SnO _x S _γ	Sn(dmamp) ₂	H ₂ O	H ₂ S		Ref. 553
SnS	Sn(acac) ₂	H ₂ S			Ref. 554
SnS	Sn(ⁱ PrAMD) ₂	H ₂ S			Ref. 555
SnS	Sn(NMe ₂) ₄	H ₂ S			Ref. 556
SnS	Sn(η ² -((N ^t Bu)CHMeCHMe(N ^t Bu))) ₂	H ₂ S			Ref. 227
SnS	Sn(dmamp) ₂	H ₂ S			Ref. 557
SnS	Sn(ⁱ PrFMD) ₂	H ₂ S			Ref. 558
SnS	Sn(η ² -((N ^t Bu)CMe ₂ CH ₂ (N ^t Bu)))	H ₂ S			Ref. 559
SnS ₂	Sn(NMe ₂) ₄	H ₂ S			Ref. 172
SnS ₂	Sn(OAc) ₄	H ₂ S			Ref. 560
SnSe	SnCl ₄	Se(SiEt ₃) ₂			Ref. 561
SnSe _x	Sn(NMe ₂) ₄	Se(SiMe ₃) ₂	NH ₃		Ref. 562
SnSi _x O _γ	Sn(N(SiMe ₃) ₂) ₂	O ₃			Ref. 536
SnTe	Sn(NMe ₂) ₄	(Me ₃ Si) ₂ Te	NH ₃		Ref. 563
SnTi _x O _γ	Sn(acac) ₂	Ti(NEt ₂) ₄	O ₃		Ref. 564
SnTi _x O _γ	Sn(dmamp) ₂	Ti(O ⁱ Pr) ₄	H ₂ O		Ref. 565
SnTi _x O _γ	Sn(NMe ₂) ₄	Ti(NMe ₂) ₄	H ₂ O		Ref. 566
SrB _x O _γ	SrTp ₂	H ₂ O			Ref. 567
SrCO ₃	Sr(Cp ^t Bu) ₃	H ₂ O			Ref. 568
SrCo _x O _γ	Sr(thd) ₂	Co(acac) ₃	O ₃		Ref. 569
SrNb _x O _γ	Sr(ⁱ Pr ₃ Cp) ₂	Nb(N ^t Bu)(NEt ₂) ₂ (Cp)	O ₃		Ref. 414
SrO	Sr(Cp ^t Bu) ₃	H ₂ O			Ref. 568
SrO	[Sr(demamp)(thd)] ₂	O ₃			Ref. 570
SrSnO ₃	Sr(ⁱ Pr ₃ Cp) ₂	Sn(NMe ₂) ₄	H ₂ O		Ref. 62
SrTi _x O _γ	Sr(Cp ⁱ Pr) ₃	Ti(O ⁱ Pr) ₂ (thd) ₂	H ₂ O	O ₃	Ref. 571
SrTi _x O _γ	Sr(Cp ⁱ Pr) ₃	Ti(NEtMe) ₄	O ₃		Ref. 572
SrTi _x O _γ	Sr(Cp ^t Bu) ₃	Ti(OMe) ₄	O ₃		Ref. 420
SrTi _x O _γ	Sr(Cp ⁱ Pr) ₃	Ti(CpMe ₅)(OMe) ₃	H ₂ O	O ₃	Ref. 573
SrTi _x O _γ	Sr ₂ (thd) ₂ (demamp) ₂	Ti(CpMe ₅)(OMe) ₃	O ₃		Ref. 570
SrZrO ₃	Sr(Cp ⁱ Pr) ₃	Zr(NMe ₂) ₄	H ₂ O		Ref. 574
Ta ₂ O ₅	Ta(N ^t Bu)(ⁱ PrAMD) ₂ (NMe ₂)	H ₂ O			Ref. 575
Ta ₂ O ₅	Ta(N ^t Bu)(NEt ₂) ₃	H ₂ O			Ref. 576
Ta ₂ O ₅	Ta(N ^t Bu)(NEt ₂) ₃	O ₃			Ref. 576
Ta ₂ O ₅	Ta(OEt) ₅	Photons			Ref. 412

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TABLE I. (Continued.)

Material	Reactant A	Reactant B	Reactant C	Further reactants	First Ref
Ta ₂ O ₅	Ta(OEt) ₅	O ₃			Ref. 303
TaN _x	TaCl ₅ · SEt ₂	NH ₃ + AlMe ₃			Ref. 577
TaN _x	TaCp(N ^t Bu)(NEt ₂) ₂	NH ₃			Ref. 578
TaN _x C _γ	Ta(NEt ₂) ₃ (N ^t Bu)	NH ₃			Ref. 579
TaN _x C _γ	Ta(N ^t Bu)(NEt ₂) ₃	H ₂			Ref. 580
TaSi _x O _γ	TaCl ₅	SiCl ₄	H ₂ O		Ref. 581
TaSn _x O _γ	Ta(N ^t Bu)(NEtMe) ₃	Sn(N ² ,N ³ -(N ^t Bu)CHMeCHMe (N ^t Bu))) ₂	H ₂ O		Ref. 582
TaZr _x O _γ	Ta(OEt) ₅	ZrCl ₄	O ₃		Ref. 583
TaZr _x O _γ	Ta(OEt) ₅	ZrCl ₄			Ref. 583
TbAl _x O _γ	Tb(thd) ₃	AlMe ₃	O ₃		Ref. 584
TbF ₃	Tb(thd) ₃	TiF ₄			Ref. 585
TbScO _x	Tb(thd) ₃	Sc(thd) ₃	O ₃		Ref. 586
TbTi _x O _γ	Tb(thd) ₃	TiCl ₄	O ₃	H ₂ O	Ref. 178
TbYb _x Ti _γ O _z	Tb(thd) ₃	Yb(thd) ₃	TiCl ₄	O ₃ / H ₂ O	Ref. 178
Ti	TiCl ₄	Me(SiMe ₃) ₂ CHD			Ref. 587
Ti	TiCl ₄	DHP			Ref. 587
TiAl _x C _γ	TiCl ₄	AlMe ₃			Ref. 588
TiAl _x C _γ	TiCl ₄	AlEt ₃			Ref. 589
TiAl _x N _γ	TiCl ₄	AlMe ₃	NH ₃		Ref. 590
TiAl _x O _γ	Ti(OEt) ₄	AlMe ₃	H ₂ O		Ref. 591
TiAl _x O _γ	TiCl ₄	AlCl ₃	H ₂ O		Ref. 592
TiAl _x O _γ	Ti(NMe ₂) ₄	AlMe ₃	H ₂ O		Ref. 593
TiAl _x O _γ	TiCl ₄	AlCl ₃	O ₃		Ref. 594
TiCo _x O _γ	Ti(OMe) ₄	Co(acac) ₃	O ₃		Ref. 595
TiCo _x O _γ	Ti(OMe) ₄	Co(acac) ₃	H ₂ O		Ref. 595
TiCr _x O _γ	TiCl ₄	CrO ₂ Cl ₂	CH ₃ OH		Ref. 596
TiC _x N _γ	TiCl ₄	AlH ₂ (N ^t BuCH ₂ CH ₂ NC ₄ H ₈)			Ref. 597
TiLi _x P _γ O _z	Ti(O ⁱ Pr) ₄	Li(O ⁱ Bu)	PO(OMe) ₃	H ₂ O	Ref. 598
TiMn _x O _γ	Ti(O ⁱ Pr) ₄	Mn(CpEt) ₂	H ₂ O		Ref. 599
TiN	TiCp ₂ ((ⁱ PrN) ₂ C(NH ⁱ Pr))	Only one half cycle			Ref. 600
TiN	TiCl ₄	N ₂ H ₄			Ref. 601
TiN	Ti(NMe ₂) ₄	N ₂ H ₄			Ref. 601
TiN	Ti(NMe ₂) ₄	NH ₃	electrons		Ref. 602
TiN	Ti(NEtMe) ₄	N ₂ H ₄			Ref. 603
TiO ₂	Ti(OMe) ₄	O ₃			Ref. 604
TiO ₂	TiCl ₄	MeOH			Ref. 596
TiO ₂	TiCp ₂ ((ⁱ PrN) ₂ C(NH ⁱ Pr))	Only one half cycle			Ref. 600
TiO ₂	Ti(O ⁱ Pr) ₂ (thd) ₂	O ₃			Ref. 605
TiO ₂	Ti(Cp)(CHT)	O ₃			Ref. 606
TiO ₂	Ti(O ⁱ Pr) ₂ (NMe ₂) ₂	H ₂ O			Ref. 607
TiO ₂	Ti(O ⁱ Pr) ₃ (ⁱ Pr ₂ AMD)	H ₂ O			Ref. 607
TiO ₂	Ti(O ⁱ Bu) ₄	O ₃			Ref. 608
TiO ₂	Ti(NMe ₂) ₃ (ⁱ Pr ₂ guan-NMe ₂)	H ₂ O			Ref. 609
TiO ₂	TiCl ₄	O ₃			Ref. 610
TiO ₂	Ti(NEtMe) ₃ (ⁱ Pr ₂ guan-NEtMe)	H ₂ O			Ref. 611
TiO ₂	Ti(NEtMe) ₃ (ⁱ Pr ₂ guan-NEtMe)	O ₃			Ref. 611
TiO ₂	Ti(O ⁱ Pr) ₂ (NMe ₂) ₂	O ₃			Ref. 612
TiO ₂	Ti(O ⁱ Pr) ₃ (ⁱ Pr ₂ AMD)	O ₃			Ref. 612
TiO ₂	Ti(EtCp)(NMe ₂) ₃	O ₃			Ref. 613
TiO ₂	Ti(O ⁱ Pr) ₄	Photons			Ref. 412

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TABLE I. (Continued.)

Material	Reactant A	Reactant B	Reactant C	Further reactants	First Ref
TiO _x P _γ	TiCl ₄	H ₂ O	PO(OMe) ₃		Ref. 614
TiO _x P _γ	TiCl ₄	PO(OEt) ₃			Ref. 615
TiO _x P _γ	TiCl ₄	P(OSiMe ₃) ₃ O	H ₂ O		Ref. 616
TiS ₂	Ti(NMe ₂) ₄	H ₂ S			Ref. 617
TiTa _x O _γ	Ti(O ⁱ Pr) ₄	Ta(NCMe ₂ Et)(NMe ₂) ₃	H ₂ O		Ref. 618
TiV _x O _γ	TiCl ₄	VOCl ₃	H ₂ O		Ref. 619
Tm ₂ O ₃	TmCp ₃	H ₂ O			Ref. 620
TmTi _x O _γ	Tm(thd) ₃	TiCl ₄	O ₃	H ₂ O	Ref. 178
TmYb _x Ti _γ O _z	Tm(thd) ₃	Yb(thd) ₃	TiCl ₄	O ₃ / H ₂ O	Ref. 178
V ₂ O ₃	VO(acac) ₂	O ₃			Ref. 621
VCr _x O _γ	VOCl ₃	CrO ₂ Cl ₂	H ₂ O		Ref. 622
VIn _x S _{3γ}	V(ⁱ PrAMD) ₃	In(ⁱ PrAMD) ₃	H ₂ S		Ref. 623
VIn _x S _{3γ}	V(NEtMe) ₄	In(ⁱ PrAMD) ₃	H ₂ S		Ref. 623
VN	V(NEt ₂) ₄	NH ₃			Ref. 624
VO ₂	V(NEtMe) ₄	O ₃			Ref. 625
VO ₂	VCl ₃	H ₂ O			Ref. 626
VO _x	VO(O ⁱ Pr) ₃	O ₃			Ref. 627
VO _x	V(NEtMe) ₄	O ₃			Ref. 628
VO _x	V(NEtMe) ₄	H ₂ O			Ref. 628
VO _x	V(NMe ₂) ₄	O ₃			Ref. 629
VO _x	V(NMe ₂) ₄	H ₂ O			Ref. 629
VO _x	V(ⁱ PrAMD) ₃	O ₃			Ref. 630
VO _x	V(ⁱ PrAMD) ₃	H ₂ O ₂			Ref. 630
VO _x	V(ⁱ PrAMD) ₃	H ₂ O	O ₂		Ref. 630
VO _x	V(ⁱ PrAMD) ₃	H ₂ O ₂	H ₂		Ref. 630
VS _x	V(ⁱ PrAMD) ₃	H ₂ S			Ref. 631
VW _x O _γ	V(NEtMe) ₄	W(CO) ₆	H ₂ O		Ref. 632
W	WF ₆	H (hot-wire)			Ref. 633
WC _x	WCl ₆	AlMe ₃			Ref. 634
WC _x	WCl ₆	AlH ₂ (^t BuNCH ₂ CH ₂ NMe ₂)			Ref. 635
WC _x O _γ	W(CO)(3-hexyne) ₃	^t BuNHNH ₂			Ref. 636
WN _x	W(CO) ₆	NH ₃			Ref. 637
WN _x	WCl ₅	NH ₃			Ref. 638
WN _x C _γ	WF ₆	CH ₄ + NH ₃ + H ₂			Ref. 639
WN _x C _γ	WCl ₅	^t BuNHNH ₂			Ref. 638
WO ₃	W(N ^t Bu) ₂ (NMe ₂) ₂	H ₂ O			Ref. 640
WO ₃	W(CO) ₆	O ₃			Ref. 641
WO ₃	W(CO) ₆	H ₂ O ₂			Ref. 642
WO ₃	W(CO) ₆	H ₂ O			Ref. 643
WO ₃	WO ₂ (^t BuAMD) ₂	H ₂ O			Ref. 374
WO ₃	W(N ^t Bu) ₂ (ⁱ PrAMD) ₂	O ₃			Ref. 379
WO ₃	W(N ^t Bu) ₂ (NMe ₂) ₂	NO ₂			Ref. 644
WO _x	WH ₂ Cp ₂	O ₃			Ref. 645
WS ₂	W(CO) ₆	H ₂ S			Ref. 646
WS ₂	W(N ^t Bu) ₂ (NMe ₂) ₂	H ₂ S			Ref. 647
WS ₂	W ₂ (NMe ₂) ₆	H ₂ S			Ref. 648
WS ₂	W(CO)(3-hexyne) ₃	H ₂ S			Ref. 649
WS ₂	WCl ₆	Bis(trimethylsilyl)sulfide			Ref. 650
WS ₂	W(N ^t Bu) ₂ (Me ₃ SiMe) ₂	H ₂ S			Ref. 651
WSe ₂	WCl ₅	H ₂ Se			Ref. 652
WSe ₂	WCl ₆	SeEt ₂			Ref. 653

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TABLE I. (Continued.)

Material	Reactant A	Reactant B	Reactant C	Further reactants	First Ref
Y ₂ O ₃	Y(ⁱ PrCp) ₃	O ₃			Ref. 654
Y ₂ O ₃	Y(ⁱ PrCp) ₂ (ⁱ Pr ₂ AMD)	H ₂ O			Ref. 655
Y ₂ O ₃	Y(ⁱ PrCp) ₂ (ⁱ Pr ₂ AMD)	O ₃			Ref. 175
Y ₂ O ₃	Y(ⁱ Pr ₂ guan-NMe ₂) ₃	H ₂ O			Ref. 656
Y ₂ O ₃	Y(CpBu) ₃	H ₂ O			Ref. 657
Y ₂ O ₃	Y(EtCp) ₂ (ⁱ Pr ₂ AMD)	H ₂ O			Ref. 658
Y ₂ O ₃	Y(ⁱ PrFMD) ₃	H ₂ O			Ref. 659
Y ₂ O ₃	Y(^t Bu ₂ FMD) ₃	H ₂ O			Ref. 660
Y ₂ O ₃	Y(^s BuCp) ₃	H ₂ O			Ref. 661
YAl _x O _y	Y(CpEt) ₃	AlMe ₃	H ₂ O		Ref. 662
YbEr _x O _y	Yb(thd) ₃ + Er(thd) ₃	O ₃			Ref. 663
YbMn _x O _y	Yb(thd) ₃	Mn(thd) ₃	O ₃		Ref. 357
YbTi _x O _y	Yb(thd) ₃	TiCl ₄	O ₃	H ₂ O	Ref. 178
YFe _x O _y	YCp ₃	FeCp ₂	H ₂ O	O ₃	Ref. 664
YO _x S _y	Y(thd) ₃	H ₂ S			Ref. 350
YO _x S _y	Y(MeCp) ₃	H ₂ S	H ₂ O		Ref. 665
YO _x S _y Eu _z	Y(MeCp) ₃	H ₂ S	Eu(thd) ₃	H ₂ S or O ₃ +H ₂ O	Ref. 665
YV _x Yb _y O _z	Y(thd) ₃	VO(thd) ₂	Yb(thd) ₃	O ₃	Ref. 666
Zn ₃ N ₂	ZnEt ₂	NH ₃			Ref. 667
ZnAl _x O _y	ZnEt ₂	AlMe ₃	O ₃		Ref. 668
ZnAl _x O _y	ZnEt ₂	Al(O ⁱ Pr) ₃	H ₂ O		Ref. 669
ZnAl _x O _y	ZnEt ₂	AlMe ₂ (O ⁱ Pr)	H ₂ O		Ref. 670
ZnAl _x Zr _y O _z	ZnEt ₂	AlMe ₃	Zr(NMe ₂) ₄	H ₂ O	Ref. 671
ZnCo _x O _y	ZnEt ₂	CoCp(CO) ₂	O ₃		Ref. 672
ZnCo _x O _y	ZnEt ₂	CoCp ₂	H ₂ O	O ₃	Ref. 673
ZnF ₂	ZnEt ₂	HF			Ref. 246
ZnFe _x O _y	ZnEt ₂	FeCp ₂	H ₂ O	O ₃	Ref. 674
ZnGa _x O _y	ZnEt ₂	GaMe ₃	H ₂ O	O ₃	Ref. 675
ZnGe _x O _y	ZnEt ₂	Ge(OMe) ₄	H ₂ O		Ref. 676
ZnGe _x O _y	ZnEt ₂	Ge(NMe ₂) ₄	H ₂ O		Ref. 225
ZnIn _x S _y	ZnEt ₂	In(acac) ₃	H ₂ S		Ref. 677
ZnMg _x Al _y O _z	ZnEt ₂	Mg(CpMe) ₂	AlMe ₃	H ₂ O	Ref. 678
ZnMg _x O _y	ZnEt ₂	Mg(CpEt) ₂	H ₂ O		Ref. 679
ZnNi _x O _y	Ni(^t BuAMD) ₂	H ₂ O	ZnEt ₂		Ref. 437
ZnNi _x O _y	ZnEt ₂	NiCp ₂	H ₂ O	O ₃	Ref. 673
ZnO	Zn(eeki) ₂	H ₂ O			Ref. 680
ZnO	[EtZn(damp)] ₂	H ₂ O ₂			Ref. 681
ZnO	ZnEt ₂	H ₂ O	O ₂		Ref. 682
ZnO	ZnEt ₂	H ₂ O	O ₃		Ref. 683
ZnO	ZnEt ₂	EtOH			Ref. 684
ZnO	ZnEt ₂	2-methyl-3-buten-2-ol			Ref. 684
ZnO _x B _y	ZnEt ₂	H ₂ O	B(O ⁱ Pr) ₃		Ref. 685
ZnO _x F _y	ZnEt ₂	H ₂ O	HF + H ₂ O		Ref. 686
ZnO _x P _y	ZnEt ₂	O ₃	P(OMe) ₃		Ref. 687
ZnO _x P _y	ZnEt ₂	H ₂ O	PO(OMe) ₃		Ref. 688
ZnO _x S _y	ZnMe ₂	H ₂ O	H ₂ S		Ref. 689
ZnS	Zn(thd) ₂	H ₂ S			Ref. 690
ZnS	ZnEt ₂	SMe ₂			Ref. 691
ZnS	ZnEt ₂	HS(CH ₂) ₅ SH			Ref. 692
ZnSi _x O _y	ZnEt ₂	SiH(NMe ₂) ₃	H ₂ O ₂		Ref. 693

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TABLE I. (Continued.)

Material	Reactant A	Reactant B	Reactant C	Further reactants	First Ref
ZnSn _x O _y	ZnEt ₂	Sn(O ^t Bu) ₄	H ₂ O		Ref. 694
ZnSn _x O _y	ZnEt ₂	Sn(NMe ₂) ₄	H ₂ O		Ref. 695
ZnSn _x O _y	ZnEt ₂	Sn(cyam)	H ₂ O ₂		Ref. 696
ZnSn _x O _y	ZnEt ₂	SnC ₁₂ H ₂₈ N ₂ O ₂	O ₃		Ref. 697
ZnSn _x O _y	ZnEt ₂	SnEt ₄	O ₃		Ref. 547
ZnTa _x O _y	ZnEt ₂	Ta(NMe ₂) ₅	H ₂ O		Ref. 698
ZnTi _x O _y	ZnEt ₂	Ti(NMe ₂) ₄	H ₂ O		Ref. 699
ZnTi _x O _y	ZnEt ₂	Ti(O ⁱ Pr) ₄	H ₂ O		Ref. 700
ZnTi _x O _y	ZnEt ₂	TiCl ₄	H ₂ O		Ref. 701
ZnTi _x Zr _y O _z	ZnEt ₂	Ti(NMe ₂) ₄	Zr(NMe ₂) ₄	H ₂ O	Ref. 702
ZnZr _x O _y	ZnEt ₂	H ₂ O	Zr(NMe ₂) ₄		Ref. 703
ZrAl _x O _y	ZrCl ₄	AlCl ₃	H ₂ O		Ref. 704
ZrF ₄	Zr(NEtMe) ₄	HF			Ref. 246
ZrF ₄	Zr(O ^t Bu) ₄	HF			Ref. 246
ZrFe _x O _y	Zr(thd) ₄	Fe(thd) ₃	O ₃		Ref. 705
ZrGe _x O _y	Zr(NMe ₂) ₃ Cp	Ge(O ⁿ Bu) ₄	O ₃		Ref. 706
ZrO ₂	Zr(NEt ₂) ₄	O ₂			Ref. 305
ZrO ₂	Zr(CpMe)(CHT)	O ₃			Ref. 606
ZrO ₂	Zr(NEtMe) ₂ (ⁱ Pr ₂ guan-NEtMe) ₂	H ₂ O			Ref. 707
ZrO ₂	Zr(NEtMe) ₂ (ⁱ Pr ₂ guan-NEtMe) ₂	O ₃			Ref. 707
ZrO ₂	Zr(NEtMe) ₃ (ⁱ Pr ₂ guan-NEtMe)	H ₂ O			Ref. 611
ZrO ₂	Zr(NEtMe) ₃ (ⁱ Pr ₂ guan-NEtMe)	O ₃			Ref. 611
ZrO ₂	Zr(NMe ₂) ₂ {η ⁵ :η ¹ -Cp(CH ₂) ₃ NMe}	O ₃			Ref. 708
ZrO ₂	ZrCl ₄	O ₃			Ref. 709
ZrO ₂	ZrI ₄	O ₃			Ref. 710
ZrO ₂	Zr(Cp)(^t BuDAD)(O ⁱ Pr)	H ₂ O			Ref. 711
ZrO ₂	Zr(Cp)(^t BuDAD)(O ⁱ Pr)	O ₃			Ref. 711
ZrO ₂	Zr(Me ₅ Cp)(TEA)	H ₂ O			Ref. 711
ZrO ₂	Zr(Me ₅ Cp)(TEA)	O ₃			Ref. 711
ZrO ₂	Zr(MeCp)(TMEA)	H ₂ O			Ref. 711
ZrO ₂	Zr(MeCp)(TMEA)	O ₃			Ref. 711
ZrO ₂	Zr(NMe ₂) ₄	O ₃			Ref. 712
ZrS ₂	ZrCl ₄	H ₂ S			Ref. 259

^aAbbreviations: ^tAm: *tert*-amyl, ^{acac}: acetylacetonate, ^{AMD}: acetamidinate, ^{Bu}: butyl, ^tBu: *tert*-butyl, ⁱBu: *iso*-butyl, ⁿBu: *n*-butyl, ^sBu: *sec*-butyl, ^{CHD}: cyclohexadiene, ^{Chex}: cyclohexenyl, ^{CHT}: cycloheptatrienyl, ^{cod}: 1,5-cyclooctadiene, ^{Cp}: cyclopentadienyl, ^{Cyp}: cyclopropenyl, ^{DAD}: diazabutadiene, ^{damp}: 4-dimethylamino-2-methyl-2-pentoxide, ^{dbca}: *rac*-N²,N³-di-*tert*-butylbutane-2,3-diamide, ^{demamp}: 1-[[2-(dimethylamino)ethyl](methyl)amino]-2-methyl-2-propoxide, ^{DHP}: 1,4-dihydropyrazine, ^{dmae}: dimethylaminoethoxide, ^{dmamb}: 1-dimethylamino-2-methyl-2-butoxide, ^{dmamp}: 1-dimethylamino-2-methyl-2-propoxide, ^{dmap}: 1-dimethylamino-2-propoxide, ^{DMBD}: η⁴-2,3-dimethyl-1,3-butadiene, ^{dmb}: 3,3-dimethyl-1-butene, ^{dmbs}: (*N,N*-dimethyl-3-butene-1-amine-*N*), ^{dmea}: *N,N'*-dimethylethylenediamine, ^{dmg}: dimethylglyoximate, ^{dmpd}: 2,4-dimethylpentadienyl, ^{dmopd}: 2,4-dimethyloxopentadienyl, ^{EtBen}: ethylbenzene, ^{eeki}: *N*-(2'-ethoxyethyl)-2-penten-2-on-4-imine, ^{edpa}: *N*-ethoxy-2,2-dimethylpropanamide, ^{Et}: ethyl, ^{FMD}: formamidinate, ^{fod}: 6,6,7,7,8,8,8-heptafluoro-2,2-dimethyl-3,5-octanedionate, ^{gem}: *N,N'*-di-*tert*-butyl-1,1-dimethylsilanediamide, ^{guan}: guanidine, ^{hfac}: hexafluoroacetylacetonate, ^{Hhfac}: hexafluoroacetylacetone, ^{hmbs}: hexamethyldisilazane, ⁱPrMeBen: η⁶-1-*iso*-propyl-4-methylbenzene, ^{Me}: methyl, ^{mhdsk}: 5-methyl-2,4-hexanediketone, ^{mmp}: 1-methoxy-2-methyl-2-propoxide, ^{NHC}: *N*-heterocyclic carbene, ^{OAc}: acetate, ^{od}: octane-2,4-dionate, ^{PAMD}: propylamidinate, ^{Ph}: phenyl, ⁿPr: *n*-propyl, ⁱPr: *iso*-propyl, ^{pyr}: pyrrolyl, ^{pz}: pyrazolate, ^{thd}: 2,2,6,6-tetramethyl-3,5-heptanedionate, ^{TEA}: triethanolamine, ^{THF}: tetrahydrofuran, ^{TMEA}: tris[2-(methylamino)ethyl]amine, ^{TMEDA}: *N,N,N',N'*-tetramethylethylenediamine, ^{TMM}: trimethylenemethane, ^{TMPDA}: *N,N,N',N'*-tetramethyl-1,3-propanediamine, ^{Tp}: tris(pyrazolyl)borate, ^{triaz}: triazene.

^bSome precursor names from the ALD database have formatting changes to improve clarity and systematicity.

Thermal ALD processes for elements were few before 2000, but since then, the number of processes published annually has remained constant at around 8. Quaternary materials are newcomers and appeared clearly around 2015.

C. ALD materials based on chemical composition

In the entire time span examined, 1033 of the 1711 processes (60.3%) were for oxides (Fig. 5). Chalcogenides (212) and pnictogenides (198) had the two next largest shares. Elements (171) had the

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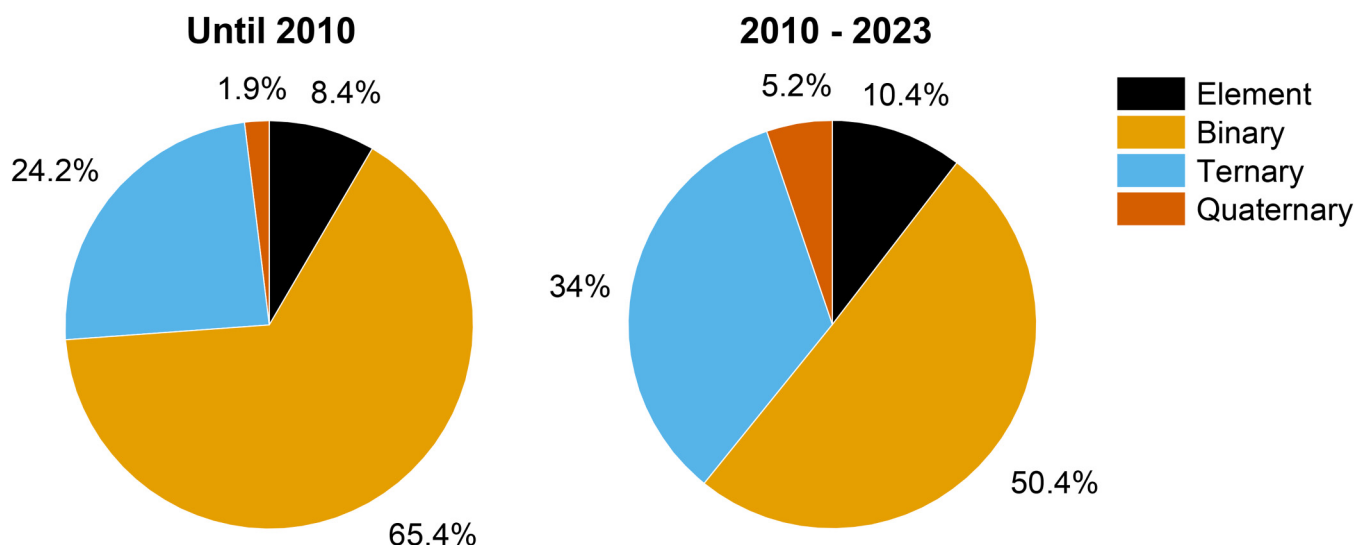


FIG. 3. Percentages of element, binary, ternary, and quaternary materials in new ALD processes before and after 2010.

fourth position, while halides (57) and carbon compounds (44) were much less studied. The comparison between the two time frames shows that the share of oxides and pnictogenides has decreased and that of elements, chalcogenides, and halides has increased. The latter reflects the more recent activity and importance of metal films, 2D materials, and halide perovskites.

When looking in more detail at the annual variation in the number of new processes, it appears, however, that the activity in

chalcogenides reached a maximum around 2017, after which the number declined (Fig. 6). In contrast, the number of halide processes has increased toward 2023. In addition, in the annual review, the number of oxide processes is declining. The curves for pnictogenides and elements appear quite even, but publishing of the element processes began years later than that of compound films.

IV. TYPE OF PRECURSORS

ALD sets many requirements for the precursors. They must be volatile, thermally stable in the deposition conditions, as well as react quickly and completely with the available surface groups and the second precursor to form the desired film composition. These requirements stem from the principles of the ALD method, but there are also other reasons for developing new precursors and processes originating from industrial applications. Reasonable price, ease of handling, lack of corrosivity, long-term stability in source temperature, and as low toxicity as possible belong to these desired properties. In recent years, the sustainability of precursors, for example, the ease of their reuse and their production emissions, has become crucial factors as well. Weber *et al.* have highlighted these requirements in their 2023 review.⁷¹³

A large variety of well-known molecules fulfill these requirements. However, in many cases, new precursors and deposition chemistries are required. Examples of the first precursor types for metals are halides, β -diketonates, and cyclopentadienyls, and for nonmetals water, hydrogen sulfide, and ozone. Alkyl silyl compounds and several heteroleptic metal complexes, such as tris(dimethylamino)cyclopentadienyl zirconium and cobalt carbonyl-*tert*-butylacetylene, have been developed for ALD applications.

ALD precursors can be grouped into 40 groups. Heteroleptic metal complexes containing at least two ligands form a large and increasing group, but they can be classified based on their individual ligands to different groups. Figure 7 shows the precursor types

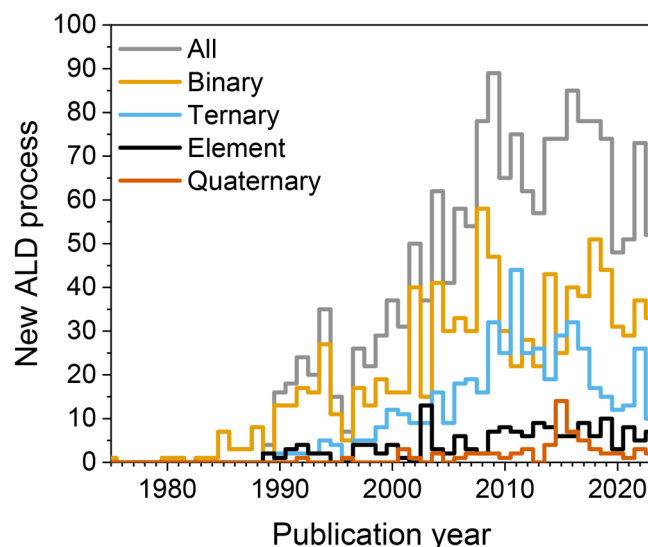


FIG. 4. Variation in the number of new processes published between 1975 and 2023.

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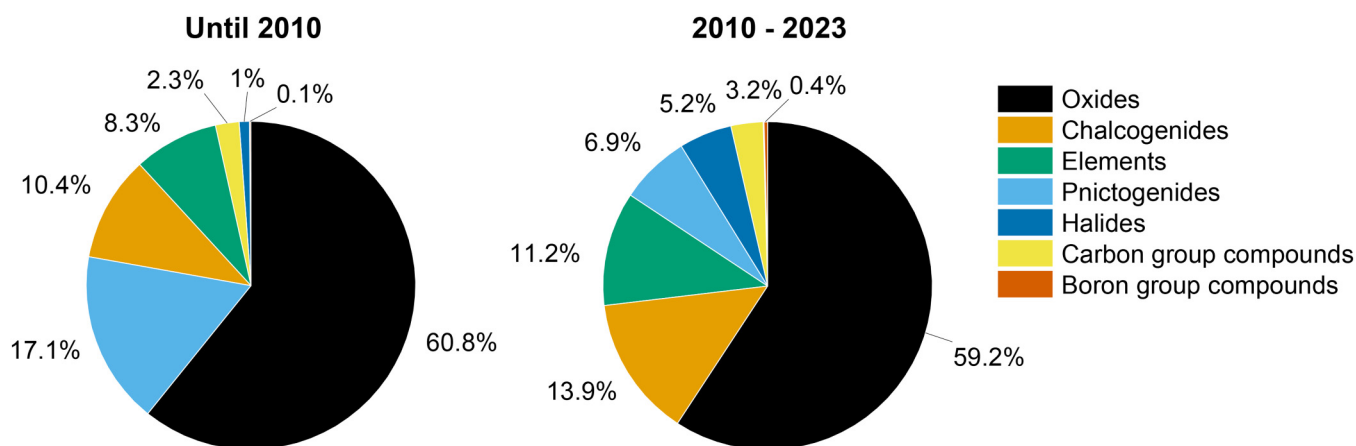


FIG. 5. Percentages of material types in new ALD processes before and after 2010.

used during the entire examined period in ALD chemistry. The 381 heteroleptic compounds are included in the individual ligand groups and thus appear two or more times in the numbers.

The order of the most applied precursor types changed over time. Before 2010, the five most common ligands used in metal precursors were halides, alkoxides, alkyls, β -diketonates, and amides/imides. The order changed after 2010 to amides/imides, unsaturated carbon ligands (mostly cyclopentadienyls), halides, alkoxides, and alkyls (Fig. 8). If the last 5 years are considered, it seems that the amount of alkoxide, cyclopentadienyl, and β -diketonate compounds in the new processes is declining, that of alkyls and halides remains constant, and that of nitrogen donor ligand-containing compounds

is increasing. Heteroleptic complexes have significantly increased in share during the last decade, becoming the most popular compound type in new processes after 2010. Among nonmetal precursors, reducing agents have become increasingly important because of the increased interest in metal deposition.

There are many reasons for the increased use of heteroleptic compounds in new ALD processes. One obvious reason is that simple homoleptic complexes were studied and applied first. The use of heteroleptic ligands within the same complex provides new possibilities for precursor development. Heteroleptic complexes can exhibit improved volatility, as seen for example in several metal halides, which change to volatile monomolecular species when

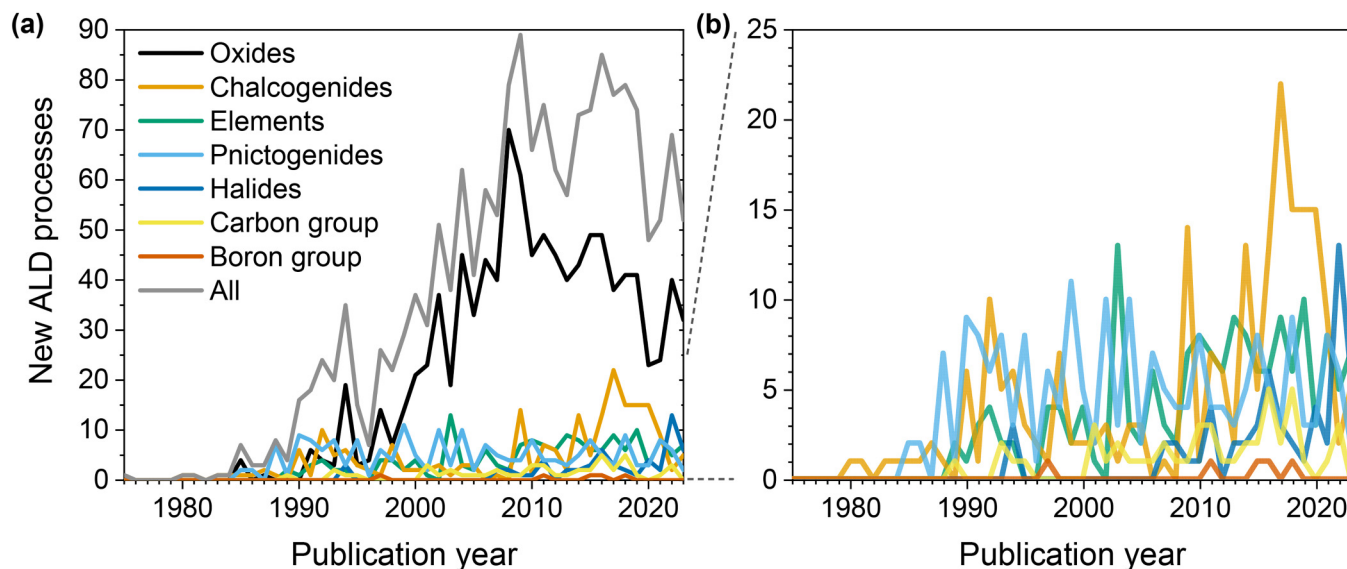


FIG. 6. Number of new ALD processes for different material types: (a) All compounds; (b) detailed picture without oxide materials.

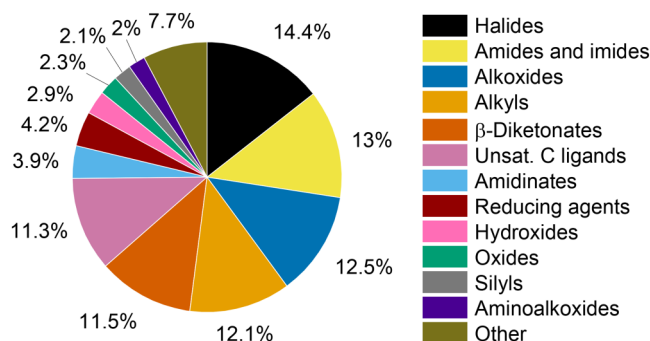


FIG. 7. Shares of most important precursor types in the entire examined period (1975–2023) used in ALD precursors. The number of compounds was 3004.

adducted by a neutral adduct molecule. It is possible to tune the reactivity and stability of the precursor using two or more different ligands. One ligand type may provide reactivity, and the other may provide stability. This is exemplified by tris(dimethylamino)cyclopentadienyl zirconium, where the amido groups react with the surface OH-groups and cyclopentadienyl yields a species stable at the surface, enabling saturative growth. Heteroleptic ligand systems can be employed to lower the melting point of the precursor so that the compound is liquid at the application temperature. Liquid precursors are desirable because they enable constant vaporization of the precursor more easily than solids, and particle problems can be avoided.

V. HIGHLIGHTS OF NEW MATERIALS

A. Elements–metals–reducing agents

ALD of elements is challenging when compared to the deposition of binaries such as oxides. Metal oxides are typically quite

stable, and the oxidation state of the metal does not necessarily need to change upon deposition. ALD of elements has steadily, albeit slowly, increased in share from the overall processes (Fig. 2). Although the development is slow, there is a clear increase in the number of new metal processes per year, averaging seven new thermal processes per year after 2010 (Fig. 9). Before 2010, the average was around four processes a year. Furthermore, after 2010, nine new elements were deposited for the first time with thermal ALD, which is almost 40% of all the elements ever deposited. This can be explained by the increasing need for ALD processes of transition metals. Besides the steady increase, one can observe a decrease in publications in 2020 caused by the COVID-19 pandemic. The year 2003 was also quite striking, but this can partially be attributed to the paper by Lim *et al.*,⁷¹⁴ in which four elements were deposited from amidates with H₂, all of them thus providing a unique process to the library.

In addition to the increase in volume, the processes have also changed in nature. Many of the processes published before 2010 utilized various metal precursors, but only a limited number of reducing agents were employed (Fig. 10). The main reducing agents used were H₂, main group hydrides, and zinc. Furthermore, singular processes can be found where NH₃,⁷¹⁵ glyoxylic acid,⁷¹⁶ formaldehyde,⁷¹⁷ formalin + ethanol,⁷¹⁸ and alcohols⁷¹⁷ were used as reducing agents. Furthermore, some processes were developed for elemental silicon and germanium that used temperature cycling, irradiation, or atomic hydrogen/hydrogen radicals.

H₂ reduction has been used for some halides, but mainly the metal precursors have been β-diketonates, acetamidates, or aminoalkoxides, where the hydrogenation of the ligand yields metallic deposit. There are also some heteroleptic precursors that work with H₂.^{142,367,486,719} The main group hydrides (mostly disilane) have been utilized for halogen stripping, yielding elemental metals from high oxidation state metal halides such as WF₆. Reduction by alcohols and formalin is achieved by oxidative dehydrogenation,^{424,720}

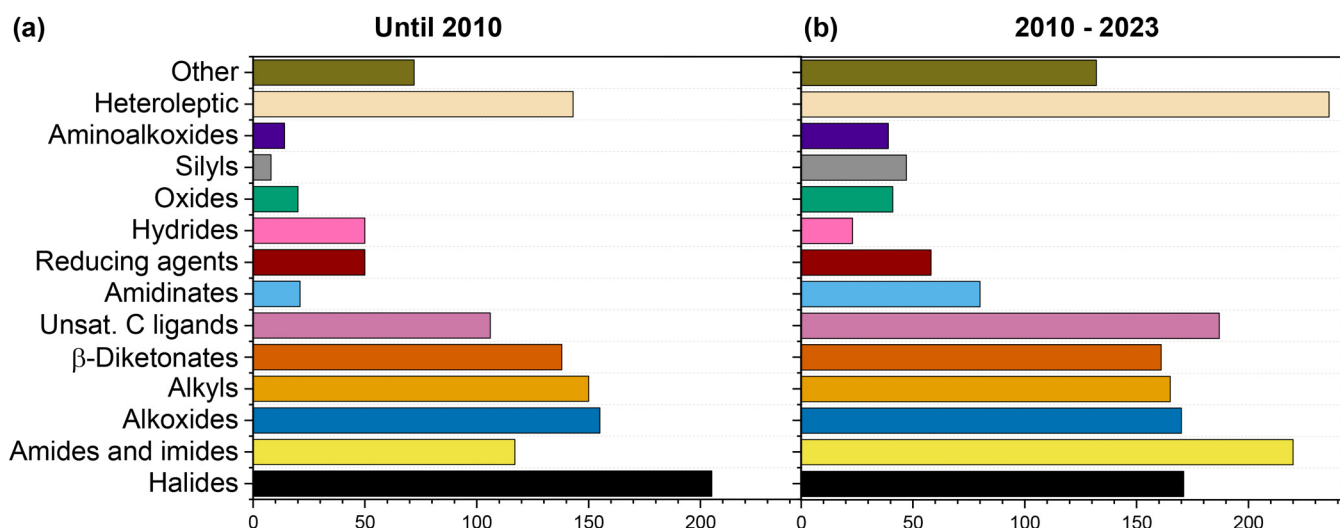


FIG. 8. Amounts of different compounds used as ALD precursors (a) prior to 2010 and (b) after 2010.

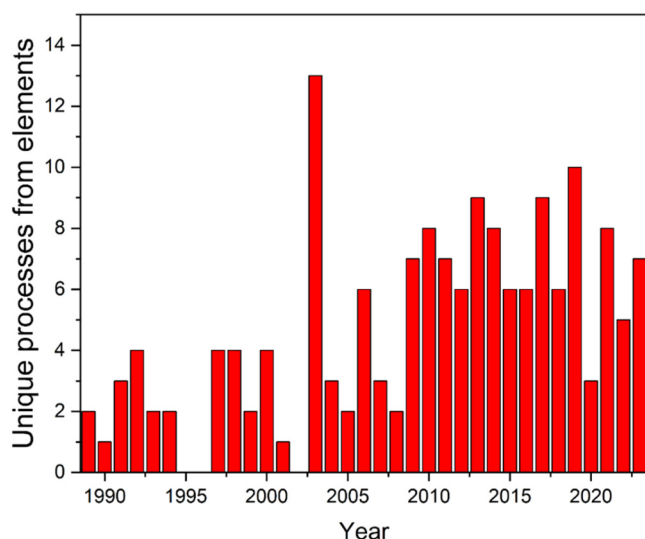


FIG. 9. Number of unique element processes published annually.

producing reactive hydrogen *in situ*. Finally, zinc and diethylzinc act as efficient reducing agents because of their redox potential or the ability of DEZ to undergo favorable ligand exchange reaction.⁷²¹ However, the use of zinc and zinc-containing compounds

as reducing agents is discouraged because of their inherent problems, such as dissolution into the film material and the risk of zinc incorporation in other process steps, which is very detrimental to the semiconductor industry.

Noble metals such as platinum group metals and gold can be prepared by ALD using oxygen or ozone as the coreactant.⁷²² The oxidative processes are enabled by the positive redox potential of the aforementioned metals and their ability to activate O_2 molecule to atomic oxygen, which oxidizes the hydrocarbons of the organometallic metal precursor.⁷²³ In oxidative processes, oxygen and ozone have been efficient reactants, and there has been no drive to find new oxidants. Molenbroek *et al.* reported the first oxidative thermal ALD process for an element, palladium, in 1998.⁷²⁴ O_2 has been in steady use both before and after 2010, whereas most processes utilizing ozone appeared after 2010. Oxidative processes exist for Ru, Rh, Pd, Os, Ir, Pt, and Au.⁷²⁵ Other metals tend to form too stable oxides, preventing the formation of metal.

In processes published after 2010, there has been a boom in not only new reducing agents but also new metal precursors. Instead of classic reductants such as H_2 , more esoteric compounds are increasingly utilized. H_2 is fairly unreactive in its molecular form and requires either high temperature or catalytic surface to work efficiently. Consequently, most of the low temperature thermal ALD processes that utilize molecular hydrogen are for catalytically active metals, such as Pd, Pt, and Ru.

The major reducing agents (used in multiple publications) introduced to ALD after 2010 are borane dimethylamine [$BH_3(NHMe_2)$],

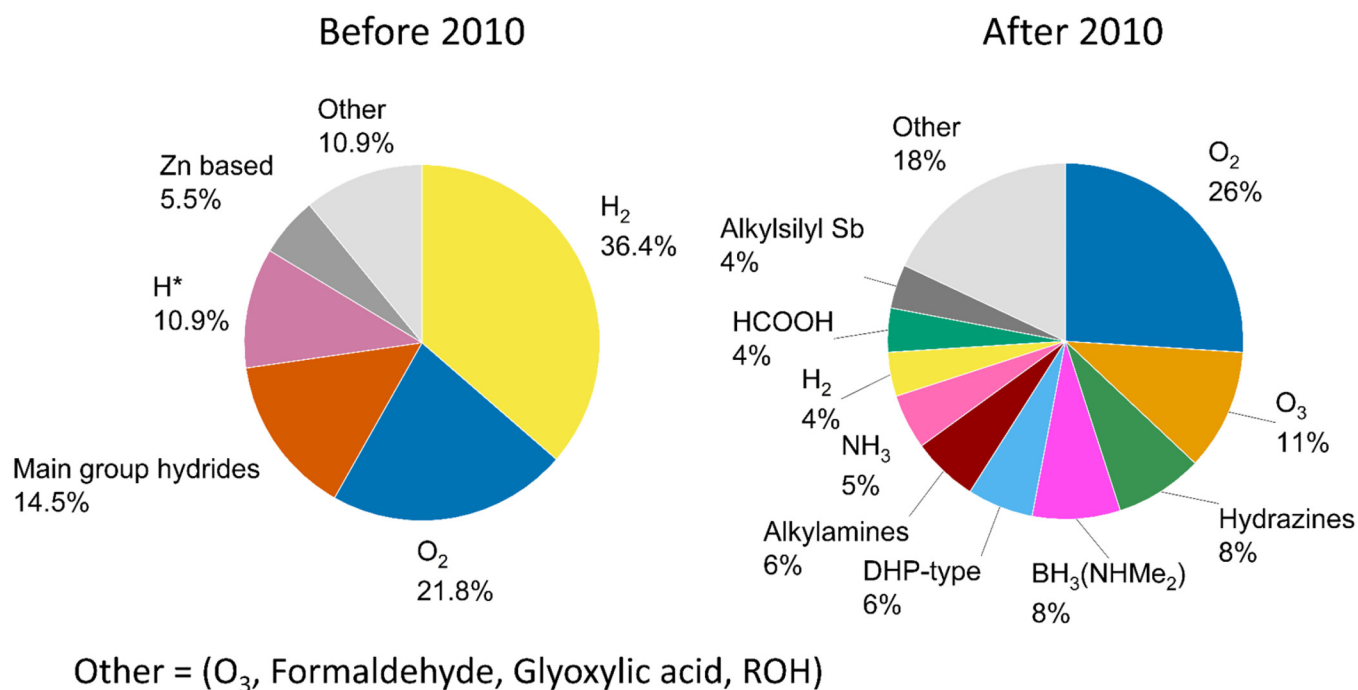


FIG. 10. Coreactants and their share in the total processes used with different metal precursors before and after 2010. H^* refers to processes that utilize atomic hydrogen or radicals.

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1,4-bis[trimethyl(silyl/germyl)]-1,4-dihydropyrazines, alkylamines, formic acid, and various hydrazines (Fig. 10). In addition, ALD of antimony was realized via a ligand exchange reaction using $(\text{Et}_3\text{Si})_3\text{Sb}$ and SbCl_3 .⁴⁸ There are also some unique processes, such as ALD of aluminum from AlCl_3 and aluminum hydride-based coreactant developed in Wayne State University.¹⁹ One should note that since ruthenium dominates the unique processes published after 2010, oxygen and ozone make up most of the coreactants.

$\text{BH}_3(\text{NHMe}_2)$ was for the first time utilized by Kalutarage *et al.*¹⁰⁵ in ALD of Co, Cr, Fe, and Ni from their respective imino alkoxides, and Cr among them was deposited for the first time by ALD. Subsequently, this reducing agent was also used to deposit Cu,¹⁴¹ Ag,¹⁶ and Au.⁵⁶ However, $\text{BH}_3(\text{NHMe}_2)$ requires either a catalytic surface (Ru) or it needs to be used to deposit an element (Ag) that can itself activate it, as it does not work as a coreactant without a presence of catalytic material.

Hydrazines and other alkyl amines besides $\text{BH}_3(\text{NHMe}_2)$ have also emerged as promising reducing agents. Hydrazines (hydrazine, dimethyl hydrazine, and *tert*-butyl hydrazine) have been used with a variety of metal precursors. Hydrazine was first used in the ALD of copper by Knisley *et al.*,¹⁴⁰ where it reduced copper formate into metallic copper. Copper formate itself was obtained by reacting $\text{Cu}(\text{OCHMeCH}_2\text{NMe}_2)_2$ with formic acid. Thus, three precursor ABC-type process could be realized. Copper was later deposited from another copper precursor, $\text{Cu}(\text{dmap})_2$, with *tert*-butyl hydrazine.¹⁴⁵ In addition, Ru,⁴⁹¹ Ag,¹⁷ Ni,⁴²⁶ and Co¹⁰⁴ have been deposited with hydrazine reduction.

Metallic cobalt can also be deposited from $\text{Co}(\text{tBu}_2\text{DAD})_2$ and formic acid without hydrazine. *Tert*-butyl amine has been used in conjunction with Co and Ni diazadienyls.^{107,425} Ammonia has also been successfully employed as a reducing agent, but it is limited to metals with unstable nitrides. Ammonia is commonly used as a coreactant in the ALD of nitrides. The effectiveness of different amines and hydrazines depends on many factors, such as the catalytic substrate, the metal precursor itself, and sometimes the presence of a third reagent.

Main group hydrides besides silanes that have had some success as reducing coreactants are $\text{AlH}_2(\text{dmetda})$, tributyl germane, and tributyl stannane. They have been employed with transition metal halides. $\text{AlH}_2(\text{dmetda})$ was successfully used for the first-time deposition of the aluminum metal.¹⁹ The latter hydrides yielded intermetallic compounds instead of pure elements.^{118,429} Nonetheless, these coreactants can still be considered reducing agents.

1,4-Bis(trimethylsilyl)-1,4-dihydropyrazine was first utilized in the ALD of Ti and Sn in the United States,^{529,587} and these were also the first elemental ALD processes for those metals. After a few years, a germanium analog of the reducing agent was first used in Finland in the ALD of Ni, Au, and Pd.^{59,428,726} These functionalized pyrazines rely on a two-part reaction where they first cleave chlorine away via the formation of volatile trimethylsilyl/germyl chlorides. Afterward, the reduction to elemental metal is achieved by aromatization of the adsorbed nonaromatic dihydropyrazine species. These reducing agents work exclusively with metal halides.

A few processes using alcohol as a reducing coreactant have also been published. Although they present some problems, such as likely carbon incorporation,⁴²⁴ one should note that metallic silver was deposited with propanol for the first time by Chalker *et al.*¹⁴

In metal processes, the metal precursors have not clearly changed in precursor types over time, but heteroleptic precursors have been used more frequently after 2010 than before. One good example is the relatively new class of metal precursors, metal halide adducts. Metal halides of early transition metals at high oxidation states are commonly used ALD precursors because of their high volatility, good reactivity, and low price. Some of them are also suitable for the deposition of metal films. However, late transition metal halides typically exist at low oxidation states and do not have the necessary volatility to be used as precursors. Recent research at the University of Helsinki has successfully integrated lower volatility metal halides as usable ALD precursors by adducting them with neutral ligands, increasing their volatility.^{128,428,430,726}

Another clear shift after 2010 in the ALD of elements is the focus on different elements (Fig. 11). Although ruthenium was quite a studied element before 2010, it accounted for an even larger share of all the unique processes after 2010 due to its importance to industry and relative ease of deposition via oxidative pathways. In addition, many processes have been published for Ni and Co, with the latter gaining almost all its processes after 2010. Although the shift is most certainly affected by the requirements of the semiconductor industry, such as the desire to replace copper in interconnects, one must also consider the novelty aspect. New processes have been published because research is always partially driven by curiosity and the search for previously undiscovered chemistries.

To conclude, a sizable number of elements have been unlocked since 2010, and the chemistry has shifted away from conventional reducing agents to more elaborate chemistries, more focused on specific metal precursor coreactant pairs instead of utilizing conventional reducing agents such as hydrogen gas. Unlocking ALD processes for new elemental metals is increasingly dependent on powerful reducing coreactants. Many novel reducing agents are nitrogen based, and amines and hydrazines make up a large fraction of these.

There are still elements that lack thermal ALD processes. Many of these are early transition metals with highly negative redox potentials. Depositing these elements via ALD would require strong reducing agents and an extremely oxygen-free environment.

B. Chalcogenides including 2D metal chalcogenides

Metal chalcogenides (sulfides, selenides, and tellurides) have a long history in ALD, starting from the first patent by Suntola in 1974, where elemental Zn and S were used to deposit ZnS .^{6,7} ZnS played a key role in the first industrial application of ALD: Electroluminescent displays. Much of the following chalcogenide research activity peaking in the early 1990s [Fig. 12(a)] was motivated by the search for alternative electroluminescent materials and epitaxial growth for (opto)electronics.⁷²⁷ After a period of somewhat lower activity in terms of new processes, a boom in ALD chalcogenide research occurred following the award of the 2010 Nobel Prize in Physics to Geim and Novoselov for the discovery of two-dimensional (2D) materials. In particular, many processes have been developed for transition metal dichalcogenides (TMDCs), 2D materials of MX_2 composition (M = transition metal or Sn; X = S, Se, and Te) that hold promise for applications ranging from electronics to energy and beyond.^{728–730} In addition, phase change

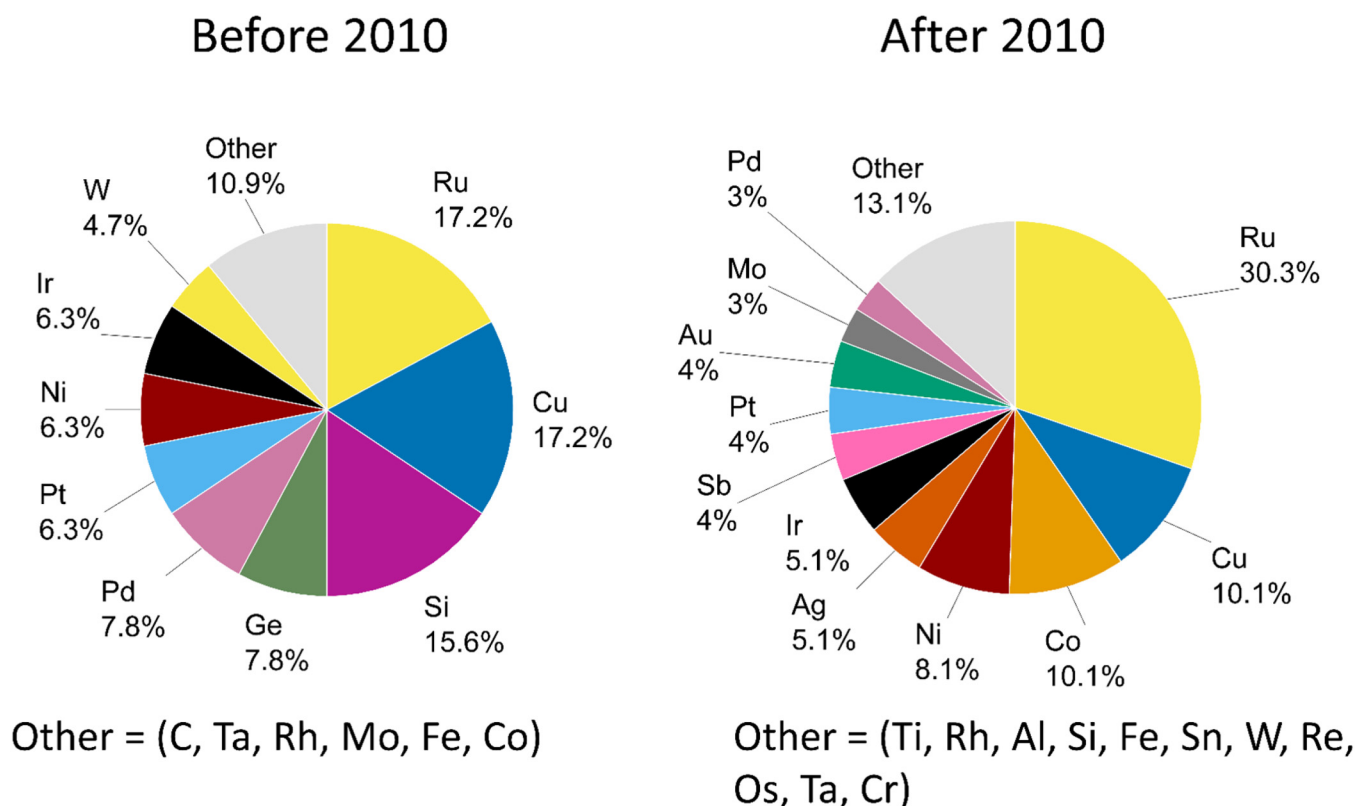


FIG. 11. Share of each element out of all thermal ALD processes before and after 2010.

memories have motivated active research on Ge-Sb tellurides in particular.⁷³¹ Furthermore, energy applications including solar cells, batteries, and (electro)catalysis have given rise to research on a range of chalcogenides, especially sulfides.⁷³² Following the peak in the mid-late 2010s (nearly 15 processes/year on average), the number of new chalcogenide processes seems to be decreasing. One reason may be the rapid expansion of the ALD chalcogenide toolbox since 2010, which enables the use of already published processes in studies. Indeed, the number of publications on ALD sulfides and TMDCs has continued to increase at least until 2020, even with the decrease in new processes.^{728,732} The toolbox of chalcogenide processes has nearly tripled since 2010, and chalcogenides have seen the second most processes in this timespan after oxides [Fig. 6(b)]. In terms of the chalcogen element, more than half of the processes deposit sulfides [Fig. 12(b)], which may be attributed to easier deposition chemistry (see below) as well as the generally higher stability of sulfides. A few mixed chalcogenide materials have also been deposited, although such ternaries are less common (10 processes) than those combining different metals (30 processes).

The material trends are illustrated in greater detail in Fig. 13. Until 2010, the most studied sulfides were those of Zn, Pb, Sr, Cu, and Cd, with 11 additional metals having at least one sulfide process developed. After 2010, molybdenum has seen the largest

number of processes (13), as MoS₂ is the prototypical TMDC material. Many processes have also been developed for 2D tungsten and tin sulfides, among others, and approximately 45% of the new sulfide processes have been for metal sulfides with 2D structures. Nonlayered sulfides, including ZnS and CuS_x, have also been actively studied. Sulfides of late transition metals and early post-transition metals are generally quite stable and thus comparably easy to deposit and stable in applications, which is reflected in their popularity. A large number of metals have seen their first sulfides deposited since 2010. Of the 2D disulfides, this includes SnS₂ (2012),¹⁷² MoS₂ (2014),³⁸⁷ ReS₂ (2018),⁴⁷⁰ ZrS₂ and HfS₂ (2019),²⁵⁹ and NbS₂ (2021).⁴¹⁹ Sulfides of a number of other metals including Li, Al, Ga, Ge, Co, Ni, Mn, As, V, Fe, and Bi have also been deposited for the first time after 2010.

The early (until 2010) research on selenides and tellurides concentrated on only a few metals, in particular, Zn and Cd (even more so if we do not consider the eight processes reported in the 2009 publication introducing alkylsilyl reagents).⁷³³ After 2010, Ge- and/or Sb-based materials largely aimed at phase change memories that have dominated the ALD selenide/telluride field.⁷³¹ Also 2D diselenides MoSe₂ (2018)⁴⁰⁰ and WSe₂ (2016)⁶⁵² and a sole 2D ditelluride MoTe₂ (2021)⁴⁰⁵ have been deposited for the first time, with further studies also emerging especially on MoSe₂. Bi₂Se₃ and

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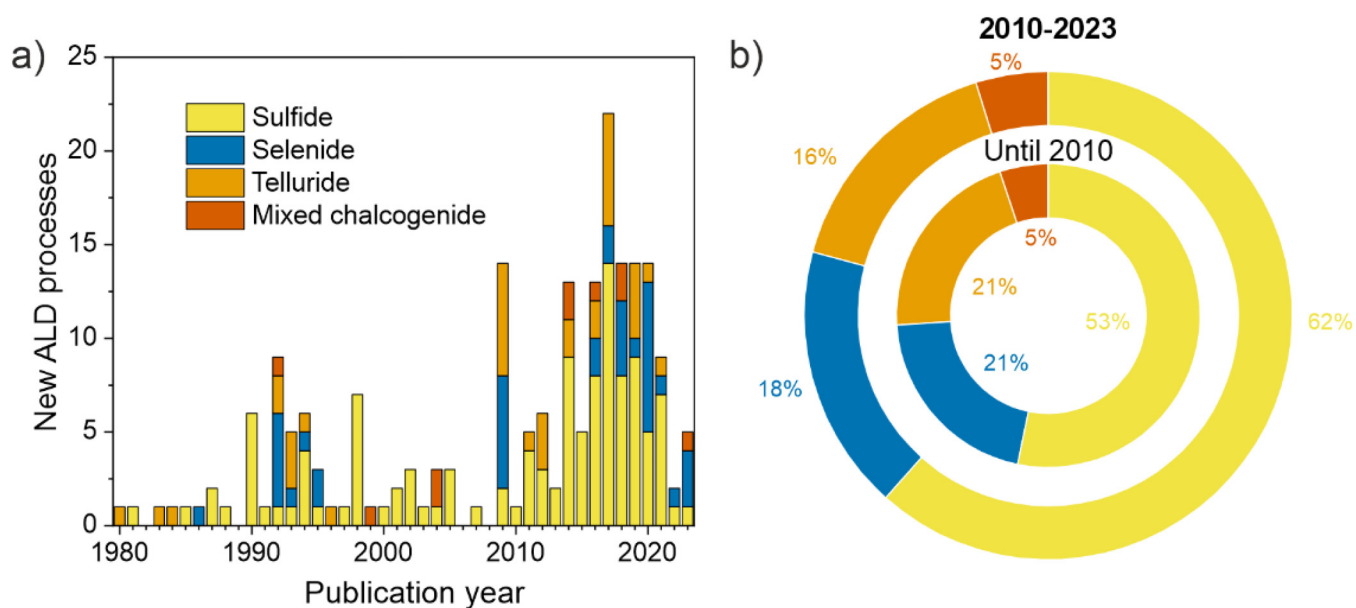


FIG. 12. (a) Annual number of new ALD processes for chalcogenides. (b) Share of different chalcogenides of processes published before and after 2010 (total number of processes until 2010: 77 and 2010–2023: 125).

Bi_2Te_3 , well-known thermoelectric materials, have also attracted attention after their first process in 2009.⁷³³ Selenides of In (2017)²⁹⁰ and Fe (2023)²⁰⁵ have also been deposited recently. Regardless, the number of metals with available selenide (11) and

telluride (10) ALD processes remains limited compared to that of sulfides.

In terms of complex chalcogenides, 10 ternary sulfides (half of them containing Cu) have been deposited for the first time since

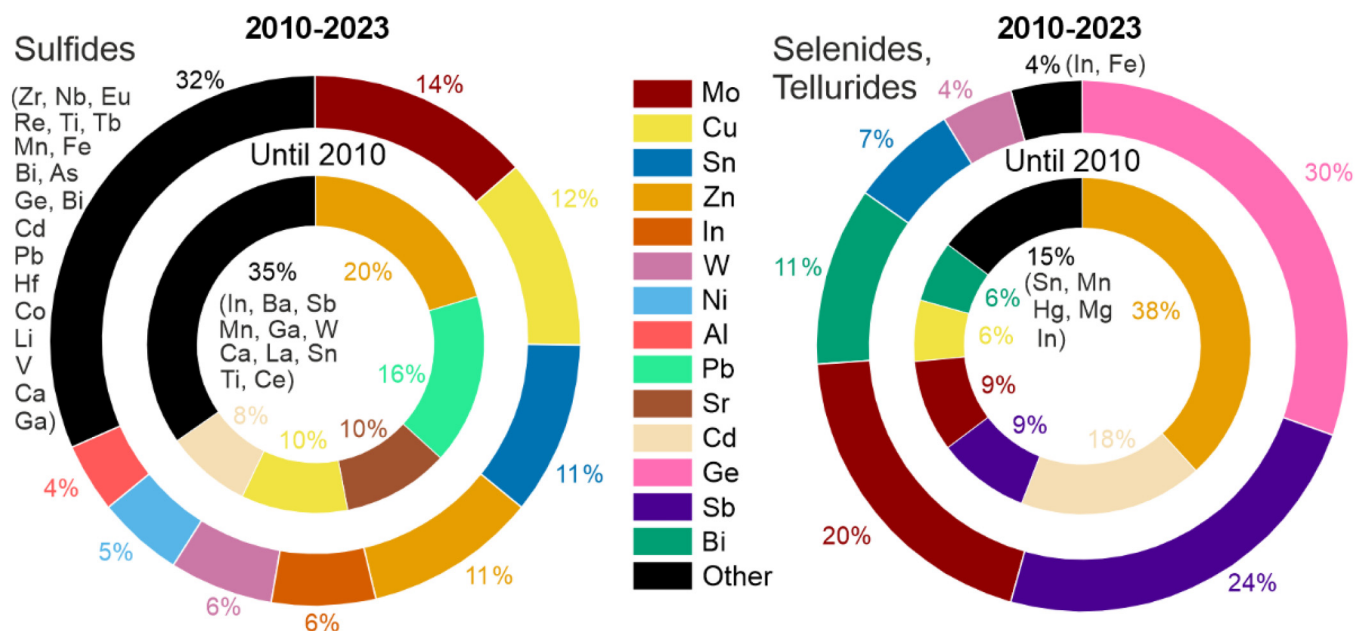


FIG. 13. Chalcogenide processes grouped by chalcogen (left: S; right: Se, Te) and metal before and after 2010. The total number of sulfide processes is 49 before 2010 and 95 after 2010; selenides and tellurides 34 before 2010 and 46 after 2010 (note that each metal in a multinary film is counted separately).

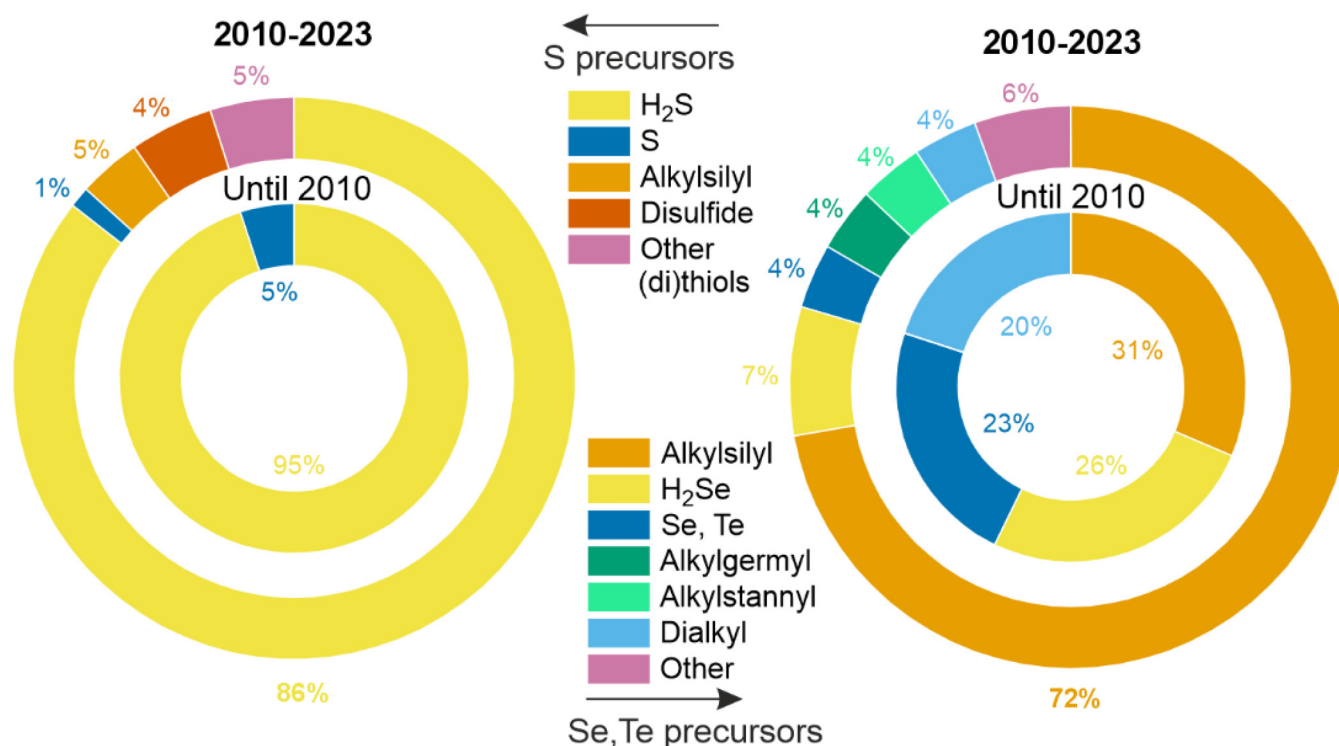


FIG. 14. Chalcogenide precursors used for S (left) as well as Se and Te (right). The total number of sulfide processes before 2010 is 45 and after 2010 is 78, while the numbers are 35 and 54 for selenides and tellurides before and after 2010, respectively (note that each chalcogen in a mixed chalcogenide is counted separately).

2010. In terms of mixed chalcogenides, three selenotellurides (first in 2014)⁹⁶ and a sulfoselenide (GeS_xSe_y)²³⁵ have been deposited. Even a few quaternary materials, including $\text{CdZn}_x\text{Ga}_y\text{S}_z$,⁹⁷ $\text{CuZn}_x\text{Sn}_y\text{S}_z$,¹⁷² and $\text{GeSb}_x\text{Se}_y\text{Te}_z$,²²⁸ have been deposited. The first two contain three metals with a single chalcogen, whereas the last one combines two metals with two chalcogens. The faster diffusion in chalcogenides compared to oxides provides both opportunities and challenges for deposition of complex materials.^{734,735}

The vast majority (~90%) of sulfide processes use H_2S as the reactant, and only a slight decrease in its use after 2010 is seen (Fig. 14). The toxicity of H_2S has motivated the search for alternative sulfur precursors, such as alkylsilylsulfide $\text{S}(\text{SiMe}_3)_2$ ³⁹¹ and disulfides MeSSMe ,³⁸⁷ EtSSEt ,³⁹⁰ and tBuSStBu .⁴⁴⁵ The reactant used in the first ALD process, elemental sulfur, also sees occasional use along with dialkylthiols and dithiols. Although some of the alternatives appear to perform well, at least for selected chemistries,⁴⁴⁵ for now H_2S remains the near universal sulfur precursor thanks to its reactivity with a broad range of metal precursors (e.g., halides, amides, and β -diketonates).

For Se, H_2Se has been used, but it has greater toxicity compared to H_2S , and for Te, the stability of H_2Te is insufficient for long-term storage and use.⁷³⁶ The elements themselves have seen some use, especially until 2010, but possess limited volatility and reactivity. The more volatile but rather unreactive dialkyl compounds have also all but fallen out of use (Fig. 14). Our group

introduced alkylsilyl reactants $\text{Se/Te}(\text{SiR}_3)_2$ in 2009.⁷³³ The volatile and reactive alkylsilyl precursors have taken over the selenide/telluride process space since then, representing over 70% of the processes published since 2010. Besides the simple and popular bis(trimethylsilyl) and bis(triethylsilyl)selenide and telluride compounds, cyclic silylselenides and tellurides have been developed at the University of Pardubice.^{401,403,405} The group has also demonstrated alkylstannyl selenide⁴⁰² and alkylgermyl telluride²³⁸ precursors, where Si is replaced by Sn and Ge. Beyond chalcogenides, at the University of Helsinki, related alkylsilyl compounds have been used to deposit antimonides,⁷³³ arsenides,²¹³ and most recently phosphides⁷³⁷ in place of the extremely toxic hydrides. Thus, the alkylsilyl compounds have broad applicability. However, as their typical reaction mechanism is driven by halosilane elimination,⁷³⁸ they tend to work the best with metal halide precursors (processes with alkoxides, alkylamides, and others also exist). In some cases, the alkylsilyl compounds have been codosed with methanol²³⁶ or NH_3 ^{232,237} to generate highly reactive $\text{H}_2\text{Se}/\text{H}_2\text{Te}$ species.

In summary, ALD of chalcogenides has experienced rapid growth after 2010, driven largely by 2D transition chalcogenides, energy applications, and phase-change memory materials. Although sulfides of 32 elements have already been deposited, many platinum group, alkaline, and rare earth metals are still lacking processes, as evident from a comparison to ALD of oxides (61 metals). Many metals form sulfides at different stoichiometries,

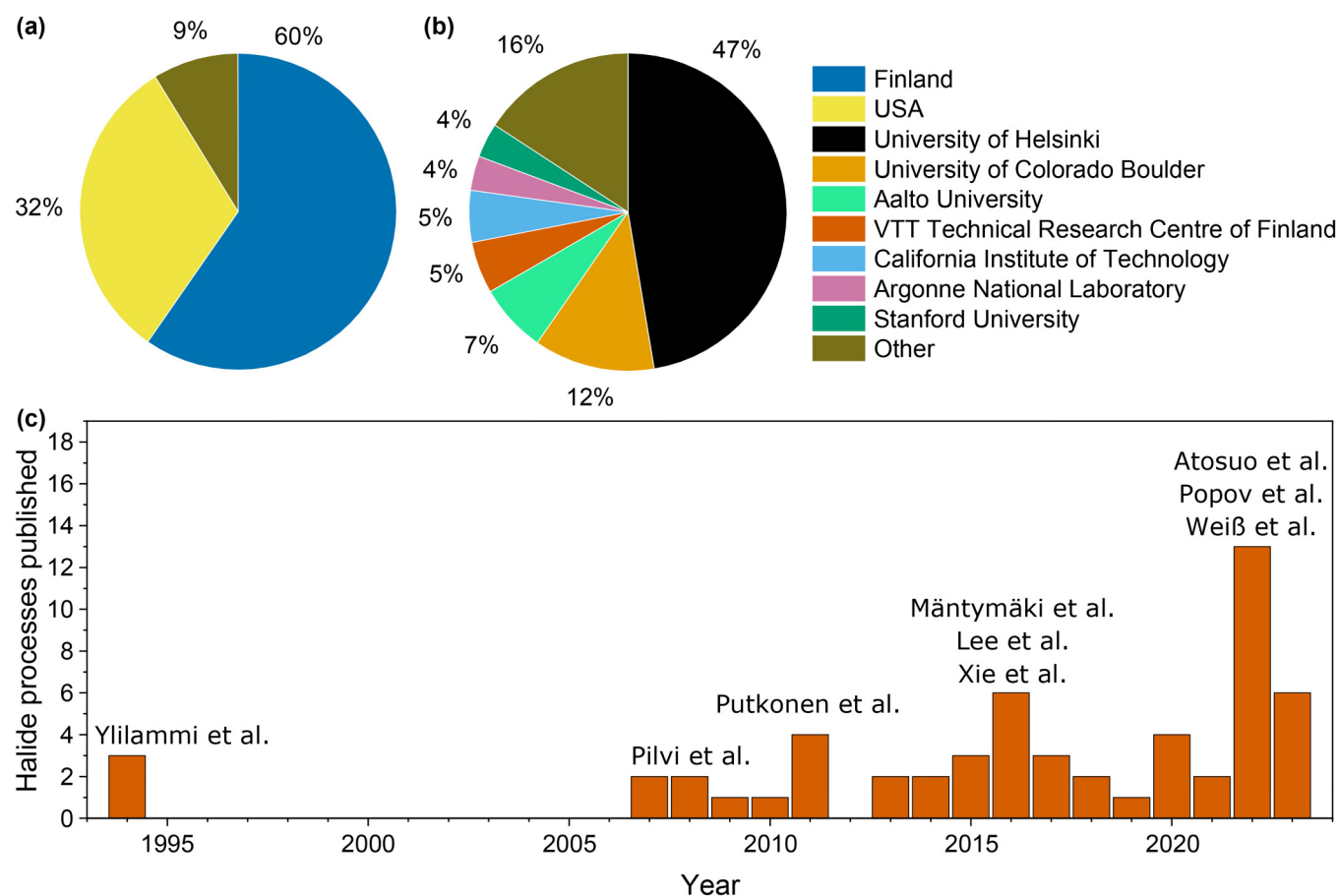
and controlling the deposited phase is largely an open question, especially without the use of plasma.⁷³⁹ Far fewer selenides and tellurides have been deposited (~10 metals). Both the studied metals and the Se/Te precursors have undergone major changes after 2009. Combining the alkylsilyl reactants with halide adducts for metals lacking volatile halide compounds is expected to play a key role in broadening the availability of ALD selenide and telluride processes. New Se/Te precursors or ways to generate highly reactive hydrides *in situ* are also promising approaches. It is evident that there is ample room for the development of new ALD chalcogenide processes. The level of activity will ultimately be affected by what materials and applications are trending. Beyond new processes, it remains to be seen when and what industrial applications of ALD chalcogenides will arise.

C. Halides

Metal halide ALD process development has deep roots in Finland. Of the 57 processes in the ALD database, 35 were

developed in Finland, of which 28 were developed at the University of Helsinki [Figs. 15(a) and 15(b)]. The United States contributes 18 entries to the total number of metal halide processes, with seven processes developed at the University of Colorado Boulder. Ylilammi and Ranta-aho published the first halide processes for CaF_2 , ZnF_2 , and SrF_2 in 1994⁷⁴⁰ almost two decades after the first oxide ALD processes in the ALD database by Hanke *et al.*⁷⁴¹ [Fig. 15(c)]. The work of Ylilammi and Ranta-aho was followed by 13 years of inactivity until the start of Dr. Pilvi's Ph.D. thesis project on ALD of metal fluorides, with the first outcome of that project published in 2007.⁷⁴² What follows is a period of activity until present times that can be traced to consistent output of Ph.D. thesis projects (for example, Mäntymäki, Atosuo, Popov, and Weiß), singular works including many processes (for example, Putkonen, Lee, and Xie) and other singular works on single processes (e.g., Hennessy).

Halide ALD processes have only a small share (~3%, Fig. 5) of the total number of ALD processes. This is undoubtedly because of the challenges of working with them. Most metal iodides,



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FIG. 15. Shares of (a) countries and (b) institutions of the halide ALD processes, based on the affiliation of the first author. (c) Number of new halide processes published per year.

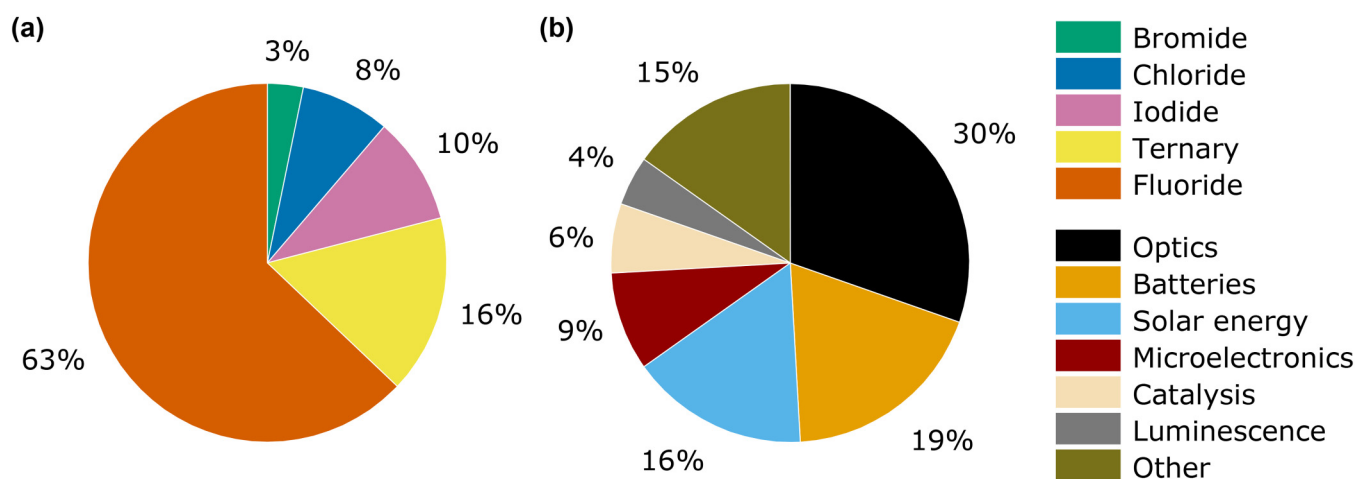


FIG. 16. Shares of different halides (a) and applications (b) in metal halide ALD processes.

bromides, and chlorides degrade in ambient air, and many also have relatively high vapor pressures. These factors make process development extremely challenging for metal iodides, bromides, and chlorides, and this is reflected in metal fluorides having the most ALD processes [Fig. 16(a)]. The dominance of fluorides is also visible in the distribution of applications of metal halide ALD processes [Fig. 16(b)]. Applications with the largest share, for example, optics and batteries stem from metal fluorides with many processes available, such as LiF (7), CaF₂ (4), MgF₂ (4), and AlF₃ (3) processes. Several metal fluorides such as CaF₂ and MgF₂ are known to possess some unique optical properties such as low refractive indices and wide regions of transparency down to deep UV regions, whereas several ternary lithium fluorides such as Li₃AlF₆ are notable ion conductors applicable in lithium-ion batteries.

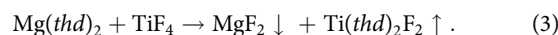
Only a few metal halide processes have been published before 2010. With the exception of the first metal chloride process for CuCl in 2010 by Natarajan *et al.*,¹⁵⁶ all processes published up to 2019 were metal fluorides. In 2019, Popov *et al.* published the first process for a metal iodide PbI₂,⁴⁵⁰ and later in 2022, the first process for a metal bromide PbBr₂.⁴⁴⁸ Since 2019, the number of iodide, bromide, and chloride processes has steadily increased owing to the interest in halide perovskites. Binary metal iodides, bromides, and chlorides are prerequisites for ternary halide perovskites. Halide perovskites are a group of compounds with remarkable optoelectronic and defect properties.⁷⁴³ They are most famous for their photovoltaic performance but are also considered in many other applications (Fig. 17). The interest in halide perovskites is visible in solar energy and microelectronics being the third and fourth biggest applications for halide ALD processes [Fig. 16(b)], as well as ternary halides and iodides being the second and third most abundant [Fig. 16(a)]. This is because the best-performing halide perovskites are ternary metal iodides such as CsPbI₃.

The chemistry of metal halide ALD processes can be categorized into three groups (Fig. 18). In the oldest group, a metal precursor reacts with a corresponding hydrogen halide [Eq. (1)].

Therefore, the oldest fluoride processes were developed using HF gas [Eq. (2)].⁷⁴⁰ Later, also HF-pyridine³³³ and NH₄F³³⁶ were also employed, but these precursors rely on producing HF and thus the same chemistry. HF is a highly poisonous and corrosive gas that reacts with many common metals used in ALD tools. For these reasons, alternative fluoride precursors were developed, and later, the same principles were transferred to other halides as well,



The first alternative to HF was developed in 2007 by Pilvi *et al.*⁷⁴² TiF₄, a volatile metal fluoride, was used as a fluorine precursor. TiF₄ undergoes ligand exchange with Mg(thd)₂, forming the desired MgF₂ and volatile Ti(thd)_xF_{4-x} species [Eq. (3)]. In later works, the family of volatile metal halide compounds as halogen precursors was expanded to include also TaF₅,⁷⁴⁵ SnI₄,⁴⁵⁰ SnBr₄,⁴⁴⁸ SnCl₄,⁴⁴⁸ TiBr₄,⁴⁴⁸ GaCl₃,⁴⁴⁸ and NbF₅.¹²³ Roughly half of all halide ALD processes employ these volatile metal halide compounds as halogen precursors (Fig. 18) because they are easier to obtain and safer to use. However, this comes at the cost of possible impurity metal incorporation into the desired compound, which can be detrimental to application-relevant properties,



The third chemistry with the smallest share of processes (Fig. 18) relies on the combustion of a halide-containing compound such as hfac (hexafluoroacetylacetone) with O₃. Fluorine from hfac forms fluoride with a metal from a metal precursor delivered previously. Any remaining precursor ligands are also combusted. It is also possible to deliver hfac ligands and a metal with

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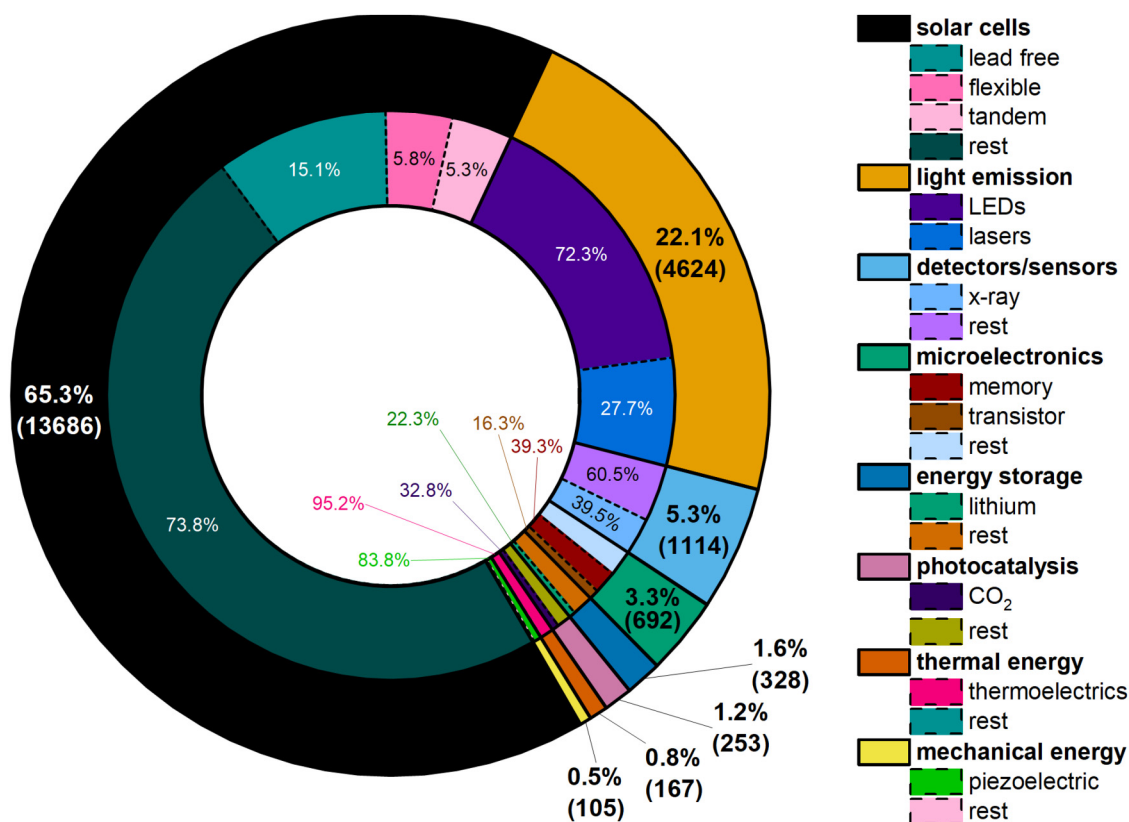


FIG. 17. Number of publications mentioning specific applications of halide perovskites. Copyright © Popov CC BY 4.0. Reproduced with permission from Popov, *Thin Film Deposition Processes for Halide Perovskites* (University of Helsinki, 2023) (Ref. 744). Copyright 2023 by G. Popov.

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the same precursor, such as $\text{Ca}(\text{hfac})_2$. This approach was pioneered by Putkonen *et al.* in 2011.⁸⁸

In terms of metal precursors, metal β -diketonates and amides are the two largest groups, accounting for 2/3 of all metal precursors in halide ALD processes (Fig. 19). β -diketonates are the most common choice of metal precursors for fluorides in combination with HF,

whereas for iodides, bromides, and chlorides, metal amides dominate as metal precursors because they typically have lower source temperatures and allow lower deposition temperatures at which the desired metal iodides, bromides, and chlorides do not volatilize.

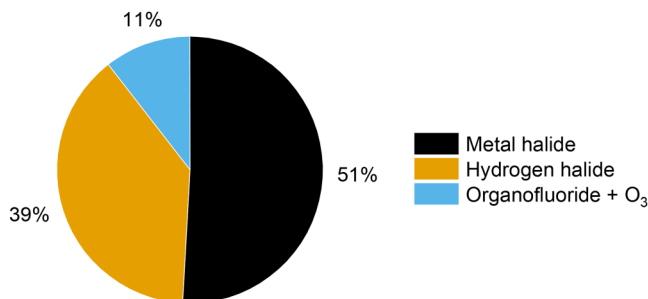


FIG. 18. Shares of different halogen precursors in the metal halide ALD processes.

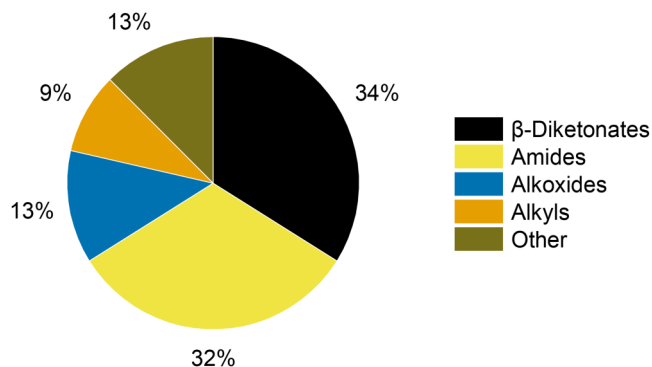


FIG. 19. Shares of different metal precursors in the metal halide ALD processes.

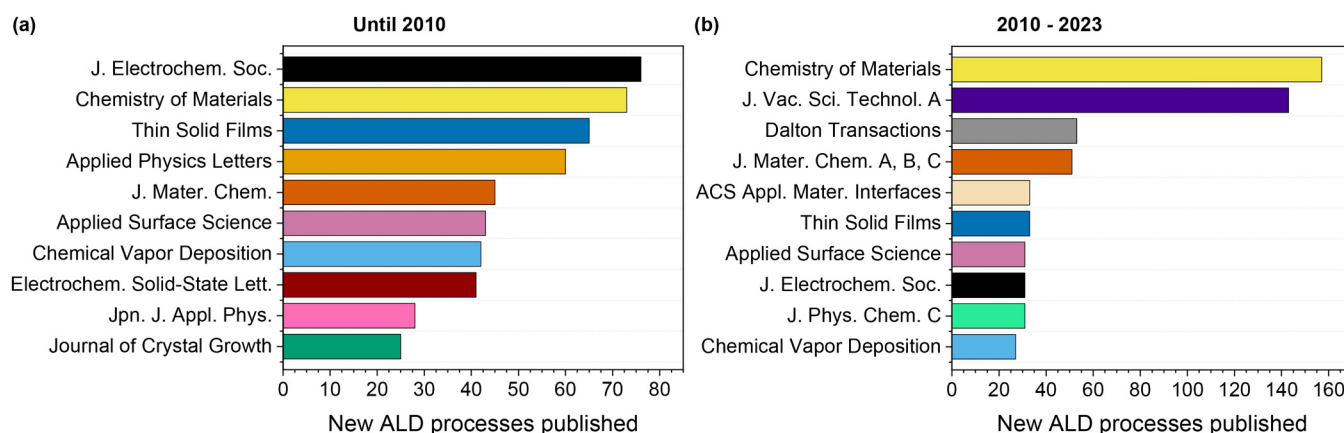


FIG. 20. Number of papers dealing with new ALD processes in the top 10 journals (a) between 1975 and 2010 and (b) since 2010. Note the scale differences.

VI. JOURNALS

New ALD processes have been reported in 170 different journals (1975–2023). The list of the top 10 journals has changed significantly over time. Only Chemistry of Materials has maintained its position in the top two (Fig. 20). *Applied Physics Letters*, *Electrochemical and Solid State Letters*, *Japanese Journal of Applied Physics*, and *Journal of Crystal Growth* have disappeared from the top 10 list of papers published after 2010. New journals in the list are as follows: *Journal of Vacuum Science and Technology A*, *Dalton Transactions*, *ACS Applied Materials & Interfaces*, and *Journal of Physical Chemistry C*.

There are many reasons for these changes. Publishers have reorganized their publications and established new journals. Those effects can be seen in Electrochemical Society's journals and *Journal of Materials Chemistry* (divided into A, B, and C in the beginning of 2013). VCH-Wiley stopped publishing *Chemical Vapor Deposition* as an independent journal at the end of 2015 (27 processes in 2010–2015) and linked it to *Advanced Materials Interfaces* (19 processes in 2010–2023). The journals of the American Chemical Society (ACS) have become important media for new ALD chemistry, with 30% of new processes published in ACS journals since 2010. American Vacuum Society started the annual series of ALD conferences in 2001 and soon after recommended publishing of the papers in JVST A in the first issue of the following year. Consequently, the number of ALD papers in JVST A increased markedly, and the journal is now number two in the ranking, representing 17% of processes since 2010 (143 processes).

VII. SUMMARY AND CONCLUSIONS

The survey of ALD chemistry revealed that the number of papers reporting new processes has increased rapidly since the late 1990s. After 2010, the number of processes published annually has been quite steady. During more than 50 years, 1711 processes have been reported, of which more than 50% have been published after 2010. Most ALD processes are for binary compounds, and oxides form approximately 60% of all processes. However, the number of

processes for ternary materials is increasing. Process development is clearly dependent on application demands. This can be seen in the processes for metals, chalcogenides, and halides, which draw attention in the review of processes published after 2010. Interest in these materials stems from the need for metallization in microelectronics, new 2D materials, and perovskite solar cells, respectively.

The precursors used in the new ALD processes have also changed, and their selection has widened. Since 2010, metal amides/imides have been the most commonly used homoleptic complexes. However, heteroleptic complexes containing two or more ligands have been even more popular than homoleptic amides/imides because they allow better tuning of precursor properties. The development of reducing agents for metal films has been an important direction for study owing to the increased need for new metal processes.

We expect that the development of ALD chemistry will also evolve in the future. The introduction of new chemistries would be slow and smooth, similar to that during the 2020s, and sudden changes would only occur if the industry or academia realizes a new large-scale ALD application.

SUPPLEMENTARY MATERIAL

See the [supplementary material](#) for the Credits section of the ALD database.

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Kessels. The new submissions and comments are reviewed by Joseph P. Klesko, Harm Knoops, Bart Macco, and Vincent Vandalon, and Erwin Kessels is responsible for the overall management. Jan Buiters designed the website and took care of the programming. In addition, many people have submitted processes to the ALD database. Their names can be viewed in the credits section of the database, and we are including a snapshot of that section (taken on 02.02.2025) in the [supplementary material](#).

AUTHOR DECLARATIONS

Conflict of Interest

The authors have no conflicts to disclose.

Author Contributions

Georgi Popov: Conceptualization (lead); Data curation (lead); Formal analysis (equal); Investigation (equal); Methodology (lead); Project administration (lead); Software (lead); Validation (equal); Visualization (lead); Writing – original draft (equal); Writing – review & editing (equal). **Miika Mattinen:** Formal analysis (equal); Investigation (equal); Validation (equal); Visualization (equal); Writing – original draft (equal); Writing – review & editing (equal). **Anton Vihervaara:** Formal analysis (equal); Investigation (equal); Visualization (equal); Writing – original draft (equal); Writing – review & editing (equal). **Markku Leskelä:** Conceptualization (equal); Formal analysis (equal); Investigation (equal); Project administration (equal); Writing – original draft (lead); Writing – review & editing (equal).

DATA AVAILABILITY

The data that support the findings of this study are openly available in the ALD Database at <http://doi.org/10.6100/alddatabase>, Ref. 11. The database is licensed under CC BY 4.0, see <https://creativecommons.org/licenses/by/4.0/>.

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